Zixun Zhou 2026年07月21日

# Investigate Blog 的副本

## 规则描述板块

ZIXUN ZHOU 2026年07月21日 19:48

**此paflet阅读说明书**

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## INVESGATE DIRECTION 1: BAZI

ZIXUN ZHOU 2026年07月21日 18:21

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描述该该主题是：
1.研究方向之一
2.并且现在已作废
3.时间啥时候

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ZIXUN ZHOU 2026年07月21日 18:21

**It all started with a stupid question: Neptune has no ocean — so why does it represent ocean energy?**

https://www.amazon.com/History-Western-Astrology-Ancient-Classical/dp/1441127372

This question hit me one day and felt genuinely absurd. Astrology says Neptune represents the ocean, fluidity, and unfathomable depth — but Neptune's actual composition is methane, ammonia, and icy mixtures, with surface temperatures around -214°C. Nothing remotely oceanic. So where does this "energy" come from? The answer turned out to be simple: when astronomers discovered the planet in 1846, they named it after the Roman sea god Neptune by convention. The astrological system then took that name's mythological imagery — sea god, ocean, fluidity — and packaged the whole thing as the planet's "energy attributes." That's it. No deeper reason. This made me realise something: the "planetary energies" in astrology are fundamentally mythological associations, not properties derived from astrophysics. If that planet had been named "Forest Goddess," astrologers today would say it represents nature, vitality, and green energy — and it would sound equally convincing. The system's validity doesn't depend on what the starting point is — it depends on the self-consistent elaboration that follows once the starting point is fixed. That felt like an important finding.

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**What is astrology's logic — it was never a physical system to begin with**

https://www.amazon.com/History-Western-Astrology-Ancient-Classical/dp/1441127372

After the Neptune realisation, I started asking: what is astrology's actual operating logic? What's its basis for claiming planetary positions affect personality and destiny? The answer: it has no such basis, because it was never describing physical reality in the first place — it operates a symbolic system. Western astrology's core framework was established in the Greco-Babylonian period, when planets were just moving points of light. Nobody knew Jupiter was a gas giant or Mars a red desert. What practitioners had was: visible colour (Mars appears red → blood and war), mythology (this star is named the war god Mars → its energy is war and force), and accumulated observational impressions. Once the framework crystallised, it became self-enclosed. When Uranus, Neptune, and Pluto were discovered, astrologers simply inserted the mythological imagery of their new names — the framework itself never updated. So astrology's logic chain runs: planet → corresponding mythological figure → symbolic meaning → energy attributes. Not: planet → physical properties → actual effects on human bodies. These are parallel language systems with fundamentally different starting points. Astrology never intended to derive energy

attributes from astrophysics — it uses mythological archetypes as projection tools. Once you understand this, "why does astrology say X" always has the same answer: because that's what the myth says.

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**So why does Bazi hold up better than astrology — derived from astrology's gaps**

After thinking through astrology's problems clearly, I applied the same standards to Bazi and found its underlying logic substantially more grounded — not because Bazi is necessarily accurate, but because its physical core is more “real”.
Astrology's central leap: the positions of distant planets at the moment
of birth affect your personality. No physical mechanism exists for this.
Bazi's core chain:
birth time → corresponding season and month → actual foetal developmental environment (temperature, light exposure, maternal nutrition) → neural development and temperament tendencies → behavioural patterns → life trajectory.
The first several steps in this chain have genuine biological basis. Modern research has found statistically significant correlations between birth season and personality traits. So Bazi captured a real signal: birth time genuinely carries information about developmental environment, and developmental environment genuinely affects people.
Of course Bazi then built an elaborate symbolic superstructure on this core — Heavenly Stems, Earthly Branches, Five Elements, Luck Pillars — and whether that superstructure can support the level of predictive precision claimed is a separate question. But at least its starting point is more grounded than "sea god represents the ocean." This comparison is what made me decide to investigate Bazi rather than Western astrology.

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**What is Bazi? It's not fortune-telling — it's an encoding system.**

When I first encountered Bazi, I dismissed it as superstition. But breaking it down revealed a surprisingly rigorous internal logic. The "eight characters" (八字) encode a person's birth time into Heavenly Stems and Earthly Branches across four pillars — year, month, day, and hour. These eight characters map the relationships and relative strengths of the Five Elements (Wood, Fire, Earth, Metal, Water) at the moment of birth. Crucially, Bazi doesn't jump directly from birth time to destiny — there's an entire calculation system in between: Five Element interactions, Ten Gods relationships, Luck Pillars and Annual Pillars. Each step follows defined rules. I now think of it as a multivariate modelling system built in ancient symbolic language: time coordinates go in, Five Element dynamics are processed, and personality tendencies and life trajectory come out. Whether this encoding accurately reflects reality is a separate question — but the logical architecture itself is structurally identical to modern data modelling.

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**Bazi is a multivariate system — not as crude as "born in winter = introverted"**

I initially made the mistake of thinking Bazi was just "your birth month determines your personality." Going deeper revealed something far more complex — and that complexity is precisely where its value lies. A Bazi chart has four pillars, each containing a Heavenly Stem and Earthly Branch, with hidden stems (藏干) concealed within each branch. That alone creates a system with eight to over a dozen variables. These variables interact through Five Element generating and overcoming cycles — Wood feeds Fire, Fire feeds Earth, and so on, alongside the overcoming relationships. On top of that, Luck Pillars (大运, changing every ten years) and Annual Pillars (流年) layer additional variables into the calculation. So when people say "Bazi doesn't work," the problem might not be the system itself — it might be that most practitioners lack the skill to operate a

genuinely complex multivariate model. Like a sophisticated statistical framework with sound methodology but inconsistent operators.

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**The Barnum Effect: why vague descriptions feel like they're talking about you specifically**

This is the core psychological mechanism explaining why fortune-telling, astrology, and tarot feel accurate. In 1949, psychologist Bertram Forer gave each of his students a personality test, then handed them what appeared to be a personalised analysis. Students rated these analyses as highly accurate — but everyone received the same text, which Forer had copied from a newspaper astrology column. This is the Barnum Effect: as long as a description is sufficiently vague and positive, most people will feel it describes them personally. The human brain actively searches for matches between ambiguous information and personal experience, remembers the hits, and forgets the misses (confirmation bias). This doesn't mean Bazi is fraudulent — it means that even if Bazi descriptions were entirely random, many people would still find them accurate. It's an inherent cognitive bias, not proof of Bazi's validity or its refutation. It simply explains why "does it feel accurate to users" is a terrible way to evaluate whether a system actually works.

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**Why people need prediction systems — anxiety and uncertainty are the real drivers**

A research finding struck me as particularly sharp: once you control for emotional need variables — anxiety levels, desire for control, and tolerance for uncertainty — educational background has almost zero predictive power over whether someone believes in astrology or fortune-telling. Highly educated people and credulous ones aren't that different when it comes to this. This suggests that belief in prediction systems is fundamentally not an intellectual issue but a psychological need. When people face questions like "what will happen to me" and "do I have a destiny," genuine anxiety is produced. Prediction systems — whether Bazi, astrology, or tarot — provide a structured narrative framework that transforms "the future" from chaos into something comprehensible. This function works regardless of predictive accuracy, because what it satisfies is the psychological need to feel seen, understood, and in control. Notably, usage rates surged during the pandemic, especially among young people. That's not coincidental — when external uncertainty spikes, people need some system to re-establish a sense of order.

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**The ritual of consulting a fortune-teller has value independent of accuracy**

This challenged my assumptions: sitting down with a fortune-teller creates a distinctive cognitive state — different from scrolling your phone or casual conversation. The ritual signals "this moment matters," forces you to slow down, and focuses attention on your own situation and choices. From this angle, fortune-telling functions more as a structured self-reflection tool than purely a prediction mechanism. The fortune-teller's questions — "has work been going well recently?" "any changes in relationships?" — systematically guide you through reviewing your own life, something most people never do deliberately. So regardless of what's predicted, the process itself already performs something resembling psychological counselling. This connects to Jung's analysis — he believed astrology operates not through physical causation but as a "synchronicity" tool, helping people draw on collective unconscious archetypes to understand their situation. Whether or not you buy Jung, this functional reading aligns with what I've observed.

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**EYES 1: The illusion of control: why "knowing your fate" paradoxically makes people feel more in control**

📎 Sosis, R.H., Strickland, B.R., & Haley, W.E. (1980). "Perceived locus of control and beliefs about astrology." *Journal of Social Psychology*, 110(1), 65–71. → https://www.tandfonline.com/doi/abs/10.1080/00224545.1980.9924223
📎 Wunder, E. (1998). "Belief in astrology as a strategy for self-verification and coping with negative life-events." *European Psychologist*, 3(3), 202–208. → https://econtent.hogrefe.com/doi/abs/10.1027/1016-9040.3.3.202

This is the most counter-intuitive finding in the psychological layer. Logically, "your destiny is fixed" should induce feelings of helplessness — but research shows the opposite: people who believe in fate-based systems often score higher on certain measures of perceived control. The reason: the sense of control doesn't come from "I can change outcomes" but from "I can understand what's happening." When someone is in crisis or chaos, the worst feeling isn't the outcome itself but the bewilderment of "why is this happening" and "what comes next." Bazi or astrology provides an interpretive framework — even if symbolic and external — and as long as it transforms chaos into "readable," people experience psychological relief. One experimental study had participants learn astrology, finding that astrology students (compared to psychology or German language control groups) showed significantly increased certainty about self-descriptive attributes and heightened perceptions of having control. Not because astrology changed reality — but because it gave people a language to understand themselves.

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**EYES 2:**
**Self-concept verification: people use Bazi and astrology to confirm "who I am"**

*Psychology has a concept called Self-Verification Theory: people have a fundamental need for external feedback to be consistent with their self-concept — whether that concept is positive or negative. In other words, we don't just want to be praised; we want to be accurately seen. This mechanism is particularly visible in how people use Bazi and astrology. When someone sees a reading, the first response is often not "does this match my situation?" but a feeling of recognition: "this is describing me." Research shows astrology is especially appealing in ambiguous or uncertain situations — precisely when people most need an external system to stabilise their self-concept. This also explains why tools like MBTI, star signs, and Bazi are so popular among young people today: they offer not just "what type of person are you" but a sense of identity that has been certified by an authoritative system. "I'm an INFJ," "I'm a Scorpio," "I'm a Jiǎ Wood Day Master" — these labels also serve social functions: shortcuts for self-introduction, tools for establishing common ground with strangers.*

📎 Self-Verification Theory overview: https://www.simplypsychology.org/self-verification-theory.html
📎 Wunder (1998) — 占星作为自我验证策略的实验研究：https://econtent.hogrefe.com/doi/abs/10.1027/1016-9040.3.3.202
📎 *From Astronomy to Astrology: Testing the Illusion of Zodiac-Based Personality Prediction with Machine Learning*, arXiv 2026: → https://arxiv.org/html/2603.29033v1

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ZIXUN ZHOU 2026年07月21日 18:21

**Confirmation bias: why you only remember when it was right**

📎 Nickerson, R.S. (1998). "Confirmation bias: A ubiquitous phenomenon in many guises." *Review of General Psychology*, 2(2), 175–220. → https://doi.org/10.1037/1089-2680.2.2.175
📎 *The Barnum Effect* — The Decision Lab综述，含确认偏误与记忆选择性的讨论： → https://thedecisionlab.com/biases/barnum-effect

*This is the memory mechanism version of the Barnum Effect, worth addressing separately because it explains something very specific: why fortune-telling feels subjectively "far more accurate than it actually is." Human memory isn't a video recording — it's reconstructive and selective. When someone believes in a prediction system, the brain automatically executes two operations: first, actively searching for evidence that matches the prediction (the hits); second, reducing attentional weight on non-matching*

*evidence and forgetting it more rapidly. This is confirmation bias operating at the memory level. So a fortune-teller says ten things, three vaguely match, seven completely miss — you remember the three, you tell your friend "she was so accurate." You're not lying; this is how the brain normally operates. Research shows information consistent with existing beliefs is selected and remembered at nearly twice the rate of inconsistent information. This mechanism makes "word-of-mouth validation" of Bazi or astrology almost entirely unreliable as a system evaluation method — even if the system is completely invalid, transmitted accuracy rates will be inflated.*

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**External locus of control and fate belief: who is most likely to believe in fortune-telling**

*Not everyone is equally receptive to fortune-telling. Psychological research has identified a stable correlation: external locus of control personality orientation is significantly positively correlated with fortune-telling belief. "Locus of control" is a personality psychology concept proposed by Julian Rotter in 1954: internally oriented people believe their own actions and abilities determine life outcomes; externally oriented people believe outcomes are primarily controlled by external forces (fate, powerful others, chance). Research shows that the stronger someone's external orientation, the more they believe in and actively use astrology or fortune-telling. But there's something subtle here — this doesn't mean external locus of control is a defect. It more likely reflects the fact that when people genuinely feel "many things are outside my control," turning to external interpretive systems is a natural response. The surge in astrology use during the pandemic, the consistently higher usage rates among women (who historically have had less structural power in society) — all of this aligns with this mechanism. This isn't superstition; it's an adaptive response to specific circumstances.*

📎 Sosis, R.H., Strickland, B.R., & Haley, W.E. (1980). "Perceived locus of control and beliefs about astrology." *Journal of Social Psychology*, 110(1), 65–71. → https://www.tandfonline.com/doi/abs/10.1080/00224545.1980.9924223
📎 Rotter, J.B. (1966). "Generalized expectancies for internal versus external control of reinforcement." *Psychological Monographs*, 80(1). → Wikipedia Locus of Control综述：https://en.wikipedia.org/wiki/Locus_of_control

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**ZIXUN ZHOU** 2026年07月21日 18:21

**Jung's reframe: astrology is not physical causation — it's a psychological projection tool**

*Jung was one of the few academic psychologists of the 20th century who took astrology seriously. But his defence of it was completely different from ordinary believers' — he didn't think planets physically affect people, but that astrology is a psychological projection system, a tool for accessing collective unconscious archetypes. His core concept is "synchronicity": certain events have meaningful coincidental connections without causal relationships. A planet's position doesn't cause your personality — rather, it "resonates" with your psychological state at a given moment. This resonance is meaningful and worth interpreting, but it's not a physical mechanism. More importantly, his view on archetypes: the zodiac signs and major planets have accumulated over thousands of years into archetypal symbols in the human collective unconscious — Mars doesn't represent the planet itself but the shared human psychological archetype of "aggression/drive/competition." When you "read" yourself through this system, you're borrowing an ancient symbolic language to access inner content that would otherwise be difficult to articulate directly. This framework also makes Bazi coherent: the Heavenly Stems, Earthly Branches, and Five Elements aren't physical prediction tools, but a sufficiently sophisticated symbolic language that helps you approach self-understanding from different angles.*

📎 Jung, C.G. (1960). *Synchronicity: An Acausal Connecting Principle*. Princeton University Press. ISBN: 9780691150505 → Goodreads书目页：https://www.goodreads.com/book/show/836937
📎 Jung on Astrology — Archetypes and the Psyche综述： → https://natalecho.com/learn/astrology/carl-jung-and-astrology-archetypes-synchronicity
📎 Mindberg — Jung and Astrology: Psychological Insights： → https://mindberg.org/astrology-psyche-jung/

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ZIXUN ZHOU 2026年07月21日 18:21

**The meaning vacuum: as religion declines, people still need answers to "why"**

*This is a sociologically interesting finding. One of the core functions traditional religion provides is a "why" explanation for personal experience: why did this happen to me? Does my suffering have meaning? Do I have a place in the universe? But in Western countries, and among younger generations in China, religious belief is declining rapidly. US data from 2019 shows self-identified "religious nones" rose from 17% in 2009 to 26%. A 2023 UK survey found the proportion of the public identifying as theistic fell from 75% in 1981 to 49%. The problem: religious decline doesn't mean the need for meaning disappears. Research finds that "the need for meaning" independently predicts both religiosity and belief in paranormal phenomena (including astrology and fortune-telling) — both predictions hold after controlling for other variables. In other words, Bazi and astrology are substantially filling the "meaning explanation" functional gap vacated by religion. This doesn't mean everyone using Bazi is doing religious substitution — it means that at the social structural level, when traditional meaning supply decreases, people seek it from other sources. Co-Star astrology app's growth from 7.5 million users in 2020 to 30 million in 2023 is not coincidental.*

📎 FioRito, T.A., Abeyta, A.A., & Routledge, C. (2021). "Religion, paranormal beliefs, and meaning in life." *Religion, Brain & Behavior*, 11(2), 139–146. → DOI: https://doi.org/10.1080/2153599X.2020.1824938 → Taylor & Francis全文: https://www.tandfonline.com/doi/full/10.1080/2153599X.2020.1824938
📎 Oxford Brookes University (2025). "Why is Gen Z looking to the stars for meaning?" → https://www.brookes.ac.uk/about-brookes/news/news-from-2025/05/expert-comment-why-is-gen-z-looking-to-the-stars-f

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ZIXUN ZHOU 2026年07月21日 18:21

**Astrology and Bazi as social currency — you don't have to believe it; it's a language for connection**

*This dimension is different from all the psychological need layers above — it's purely social function. Observe reality: large numbers of young people use star signs and Bazi without necessarily believing it can predict destiny. They use it more like a social shorthand language — "I'm a Scorpio" conveys far more information than a simple self-introduction, implying a personality profile and giving the other person a quick framework for understanding you. "What's your star sign?" is a low-cost icebreaker; once both parties use this language, a form of shared cultural belonging is established. Research shows TikTok has over 4.5 million videos tagged "astrology," and astrology content has permeated fashion, beauty, and brand marketing on Instagram. Psychologist Anirudh Rao notes: "Memes allow astrology to function as both identity and humour. A person might not deeply believe in star signs but still share a 'Leo confidence' meme because it feels relatable or funny. That's how astrology enters everyday conversation." Bazi's spread among young Chinese people follows similar mechanisms — "checking compatibility" (查合婚) serves the social function of creating a shared ritual for two people to examine their relationship together, not necessarily making destiny predictions. The ritual itself creates intimacy.*

📎 Newsmeter (2025). "Astrology finds new life on Instagram as millennials and Gen Z use it for identity, community." → https://newsmeter.in/whats-new/astrology-finds-new-life-on-instagram-as-millennials-and-gen-z-use-it-for-identity-community-754657
📎 Oxford Brookes (2025) — "Community and belonging: Much like religious congregations, astrology fosters strong online and offline communities." → https://www.brookes.ac.uk/about-brookes/news/news-from-2025/05/expert-comment-why-is-gen-z-looking-to-the-stars-f

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ZIXUN ZHOU 2026年07月21日 18:21

**Fortune-telling as a "permission mechanism" — finding external authorisation for decisions you've already made**

*This is the most subtle and practically interesting function — I think many people using Bazi or fortune-telling are actually doing this without necessarily being aware of it. The scenario: a person is already internally leaning toward a decision (changing jobs, ending a relationship, moving), but because the choice carries risk or would face opposition, they need some external authorisation to give themselves permission. So they consult a fortune-teller: if the result supports their inclination, they feel*

*"permitted" and can act; if not, they might find a second fortune-teller until they get the answer they want. Research confirms fortune-telling's real impact on decision behaviour. A study published in PLOS ONE found that even among people who self-identify as non-superstitious, men's financial risk-taking behaviour significantly increased after receiving a positive fortune-telling prediction — researchers explained this as the result providing psychological authorisation to "have a reason to take the risk," reducing decision anxiety. A separate qualitative study of Chinese users of AI fortune-telling tools found users divided into two types: those seeking entertainment and psychological comfort had higher acceptance; those making major life decisions treated readings as "supplementary input rather than authoritative guidance." This distinction is itself interesting: people use the "permission mechanism" differently depending on context.*

📎 Tan, X., van Prooijen, J-W., & van Lange, P.A.M. (2022). "Positive fortune telling enhances men's financial risk taking." PLOS ONE, 17(9): e0273233. → PMC全文: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9451074/
→ DOI: https://doi.org/10.1371/journal.pone.0273233
📎 "Re-Tell the Fortune so I Can Believe It": How Chinese User Communities Engage with and Interpret GenAI-based Fortune-Telling, arXiv 2025. → https://arxiv.org/pdf/2603.27784

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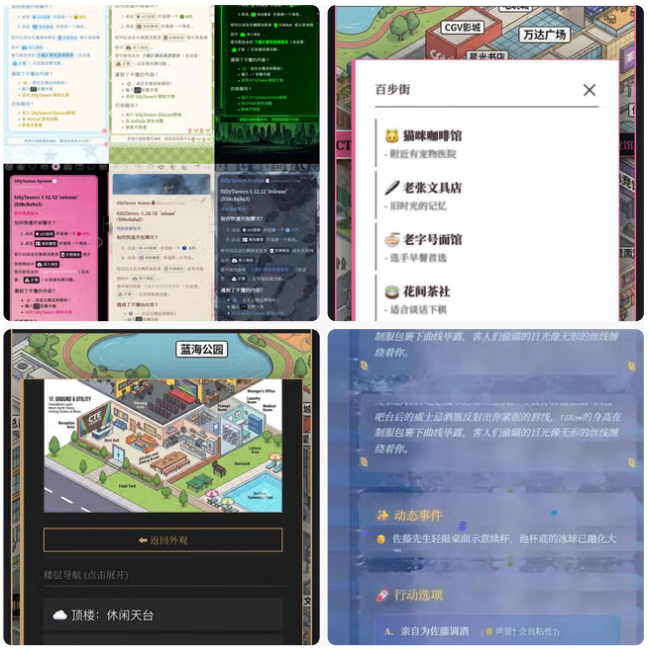
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## INVESGATE DIRECTION 2: AI ROLE PLAY

**ZIXUN ZHOU** 2026年07月21日 18:21

### WHAT IS SILLY TAVERN?

Silly Tavern is an open-source role-playing dialogue tool that allows me to engage in immersive, game-like interactions with AI characters. More than just a chat interface, it offers robust customization features, such as adjusting character personalities, building multi-character interaction scenarios, and connecting to different large language models. This tool has made me rethink the potential of "conversational creation" — instead of being a one-way creator, I now collaborate with AI to dynamically advance stories and explore characters through interactive exchanges. This collaborative creative technique has completely transformed my previous linear and isolated approach to making.

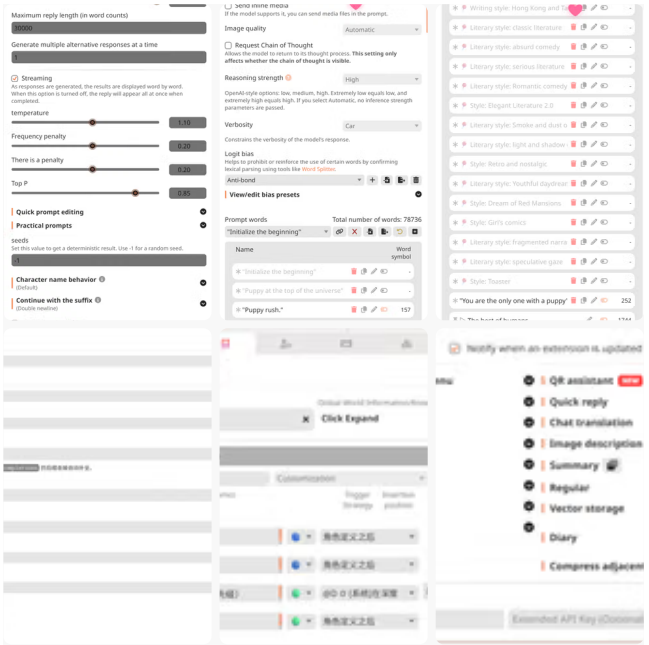



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**ZIXUN ZHOU** 2026年07月21日 18:21

### From API Configuration to Front-End Possibilities: A Tavern AI Research Log

I first came across the concept of Tavern half a year ago. I have a great passion for RPGs, D&D and wargames, so initially, I was drawn to its highly open-ended gameplay format. When I configured the API for the first time, what piqued my curiosity the most was the respective impacts of the World Book, presets, regular expressions and temperature on the AI. I could sense that their effects overlap to a large extent, much like how exposure, shutter speed and other parameters in a camera complement one another.
Thus, my first experiment was to explore how different presets influence the same AI model. I then moved on to researching the World Book, and a key insight emerged: all text-based elements – the World Book, prompts, presets, user inputs, character cards and so on – are essentially the same in nature. The only differences lie in **when** specific information is sent to the AI, the **depth** of



that information delivery, and its **position and sequence** in the process. This aspect touches on the core internal mechanics of Tavern. The division of these elements into distinct modules on the front end is merely a categorical arrangement designed to help us input the right information in the right places.
After gaining a quick and general understanding of these fundamentals, I began investigating the visual and formal possibilities that can be unlocked for AI models through prompts – in other words, the front-end content of the system.

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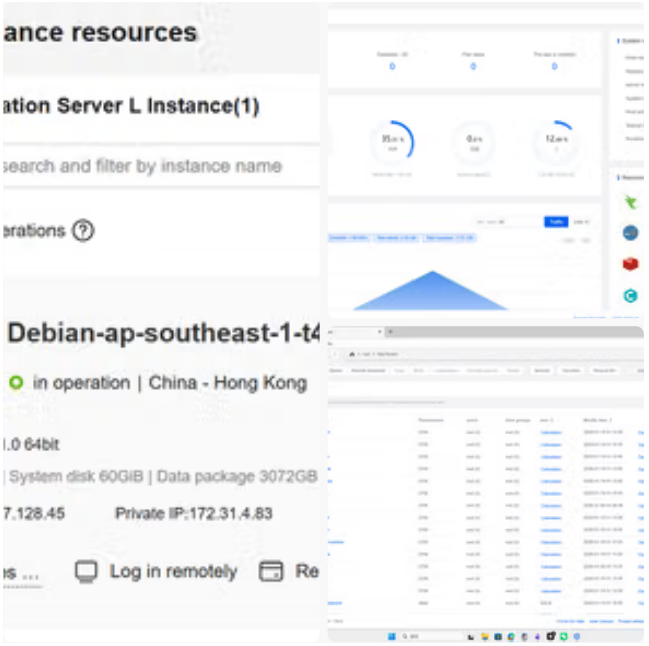
**Exploration Process of Cloud Tavern Construction**

**Exploration Direction**

Build a lightweight cloud-native Silly Tavern interactive system based on the Huawei Cloud Flexus L server and 1Panel visual panel. Explore the technical implementation and iterative problem-solving paths for cloud service deployment, visual panel operation and cross-terminal access adaptation, with a focus on the integration of technical practice and innovative application development.

**Exploration Phase**

Cloud service basic configuration → Core application deployment → Network link construction → Troubleshooting and system optimization



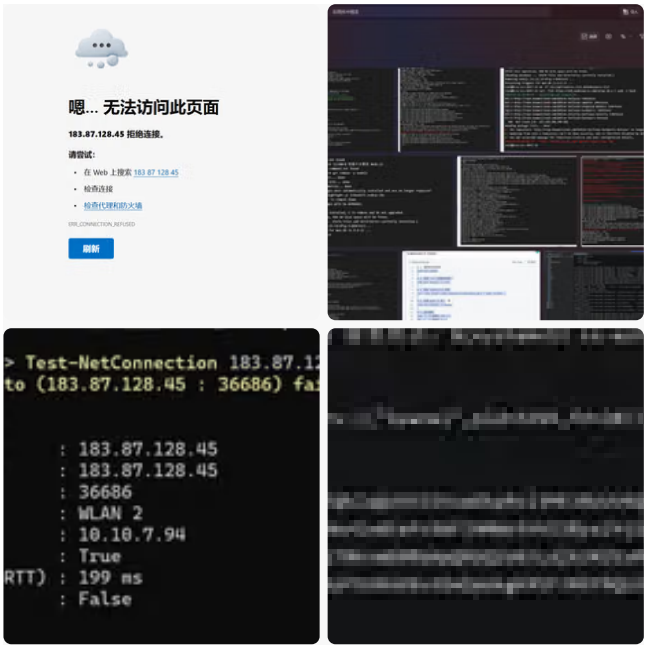

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**Network and Access Issues**

1. After the completion of the Cloud Tavern construction, external network access was unavailable. The system could only be accessed normally locally on the server or within the LAN, and the external network access link of the core application Silly Tavern was completely blocked, making it impossible to use remotely across terminals.
2. Port mapping failed after configuration. The corresponding ports have been opened in the Huawei Cloud background, but the port mapping rules configured in the 1Panel panel did not take effect. Some ports showed an open status but could not actually achieve network communication. 3. There were intermittent freezes and connection interruptions during external network access. Although the application could be accessed successfully, the stability was poor, and connection failures or excessively high response delays occurred frequently.



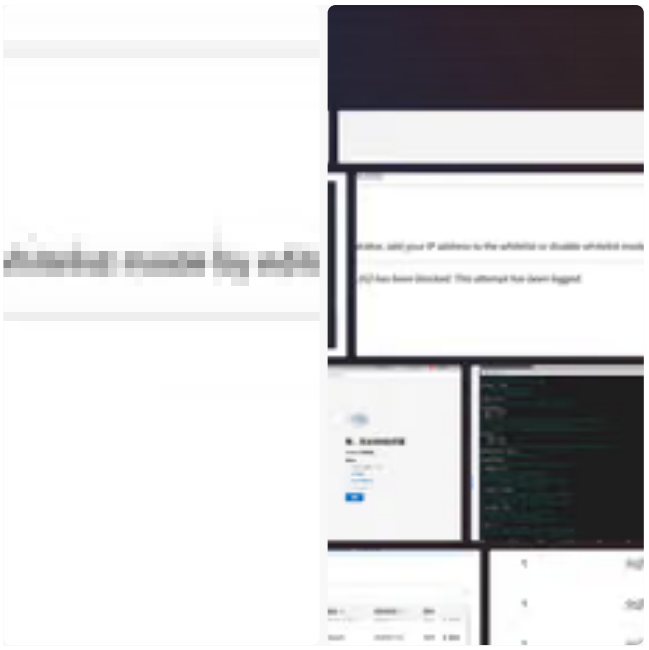

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**1Panel Panel Operation and Adaptation Issues**

1. When deploying Silly Tavern in the 1Panel panel, the application dependency installation failed. Some core plugins and runtime libraries could not be downloaded or configured normally, resulting in the failure to start the application after installation.
2. Insufficient permission errors were frequently encountered during panel operations. When modifying files and installation directories related to Silly Tavern or executing startup commands, normal operations could not be performed due to improper permission configuration, and configuration files could not be saved or take effect.
3. Unfamiliar with the operation of advanced functions of the 1Panel panel, the entrances of key functions such as reverse proxy configuration, environment variable setting, and application monitoring were difficult to locate, making it impossible to complete precise personalized configuration adjustments, which affected the efficiency of application deployment and optimization.



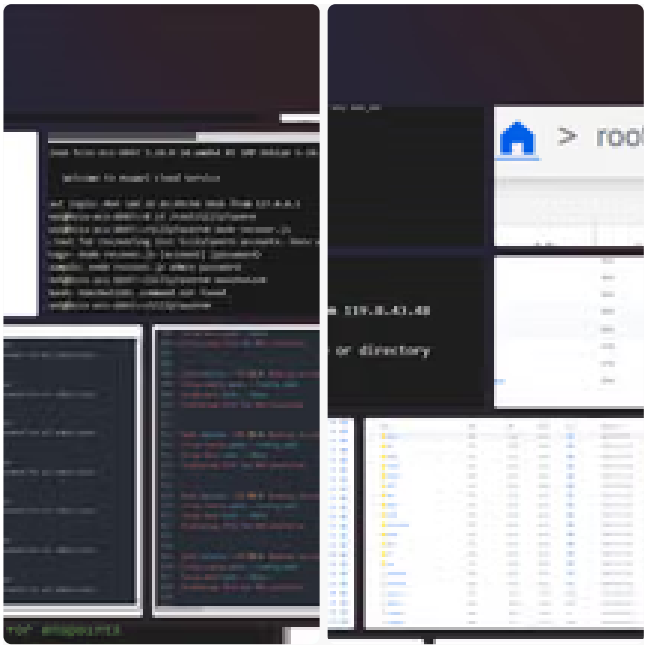
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**Silly Tavern Core Application Issues**

1. The application startup was abnormal. The 1Panel panel showed "Starting" or "Startup Failed" status, but no clear error prompt was given, or the error information was vague, making it impossible to quickly determine the core reason for the startup failure and increasing the difficulty of troubleshooting.
2. Some functional modules failed to load. The basic chat interaction function could be used normally, but extended functions (such as character library loading, conversation memory retention, and third-party plugin calls) could not run normally, and the modules got stuck or directly failed when loading.
3. There was a version adaptation conflict between the application and the server system environment. The version of the runtime environment (such as Python, Node.js)自带 by the server did not match the adaptive version required by Silly Tavern, resulting in compatibility errors or even crashes during the application operation.



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**Core Exploration Actions & Progress**

1. Completed the initial configuration of the Huawei Cloud Flexus L (monthly subscription) server and successfully installed the 1Panel visual management panel, establishing a basic operation framework for cloud services. This practice realized an efficiency upgrade from pure command-line operation to visual engineering



management, exploring a practical path for the lightweight management of cloud services.
2. Deployed the core Silly Tavern application based on the 1Panel panel, sorted out and built the basic dependency environment required for application operation, and initially constructed the core interactive function framework of the Cloud Tavern. This step achieved a key leap from basic cloud service construction to actual application implementation.




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3. Carried out the practice of building an external network access link for the Cloud Tavern, attempted to configure network parameters and open application ports on both the Huawei Cloud background and the 1Panel panel, and promoted the functional expansion from local/LAN operation to cross-terminal external network access, which exposed the core technical challenges in cloud service deployment.

4. Focused on the full-link problems in deployment and operation, and established a systematic troubleshooting thinking of "log investigation - hierarchical verification - comparison and rollback". Specifically, problems were located from multiple dimensions including 1Panel panel logs, server system logs and application local logs; the scope of problems was narrowed step by step in the order of server bottom layer → network layer → panel layer → application layer; and core technical problems such as network configuration, permission adaptation, version compatibility and resource occupation were accurately identified by comparing with official standard processes, verifying local/external network tests, and rolling back key steps.

5. Completed targeted iterative optimization for the identified problems: Restructured the port/firewall rules of the Huawei Cloud security group and the 1Panel panel to solve the problems of external network access failure and port mapping invalidation; Switched software sources, manually supplemented dependencies and adjusted file permissions to resolve dependency installation and permission errors of the 1Panel panel; Built a dedicated multi-version operating environment for the application to address the version compatibility issue between Silly Tavern and the server system; Optimized server resource configuration and application operation parameters to solve abnormal resource occupation and access stability problems, realizing a key iteration of the Cloud Tavern system from startup abnormality to basic stable operation.

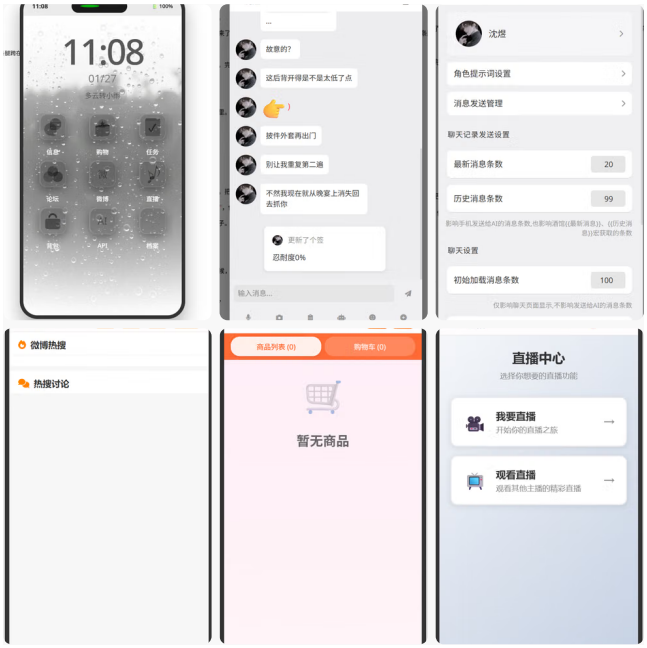
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**Floating Mobile Phone Component**

As one of the core interactive entrances, the floating mobile phone component is designed to always float on the edge of the page (drag-adjustable position), with an appearance simulating a real mobile phone interface, deeply associated with the main content of the Cloud Tavern. After clicking to expand, it can realize the integrated use of four core functional modules, fully adapting to scenario-based needs.




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1. Full WeChat functions: Support simulating friend chats (can be associated with roles in the tavern), group chat interactions, and Moments publishing (content can be synchronized with tavern dynamics). Message reminders are linked with the main content scenarios, for example, the floating mobile phone automatically pops up a prompt when receiving a role reply.

2. Music playback function: Built-in music library, supporting single playback, list loop, and volume adjustment. Background music can be switched according to the tavern scene (such as quiet chat, lively interaction), and the playback status is displayed in real time on the mobile phone floating icon.

3. Notepad function: Support quickly recording important information in the tavern (such as key points of role dialogues, scene plot nodes, to-do items). Recorded content can be exported and synchronized to the tavern personal center, and supports quick access via shortcut keys without affecting the main content browsing operation.

4. Taobao shopping software embedding: Simulate shopping scenarios, browse virtual products related to the tavern theme (such as tavern decorations, character props), support adding to cart and simulating payment processes. Shopping records can be retained in the personal center, forming a linkage with the main content plot (such as unlocking new tavern scenes after purchasing props).

5. Simulated live broadcast/watching live broadcast: Support switching between two modes - the anchor mode can simulate starting a live broadcast in the tavern (displaying scene pictures, real-time interaction), and the audience mode can watch virtual live broadcasts in the tavern (such as role special interaction, scene event live broadcast). The live broadcast comment area is interconnected with the main content chat area to enhance the scene atmosphere.

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## DEEPER INVESTIGATE - ABOUT AI

ZIXUN ZHOU 2026年07月21日 18:21

🟦**QUESTION---blue**

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🟩 **MATERIAL / PROCESS**

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🟧 **PRIMARY RESEARCH**

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🟪 **PRECEDENT**

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⬜ **CONTEXT**

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**DIRECTION1--ALIFE**

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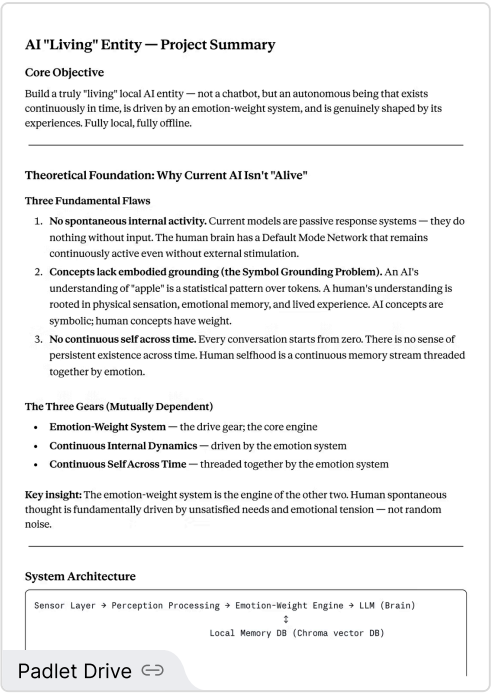
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**SUMMARY**

IDEA by Zixun

Summary by claude ops 4.6

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**AI "Living" Entity — Project Summary**

**Core Objective**

Build a truly "living" local AI entity — not a chatbot, but an autonomous being that exists continuously in time, is driven by an emotion-weight system, and is genuinely shaped by its experiences. Fully local, fully offline.

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**Theoretical Foundation: Why Current AI Isn't "Alive"**

**Three Fundamental Flaws**

1. **No spontaneous internal activity.** Current models are passive response systems — they do nothing without input. The human brain has a Default Mode Network that remains continuously active even without external stimulation.
2. **Concepts lack embodied grounding (the Symbol Grounding Problem).** An AI's understanding of "apple" is a statistical pattern over tokens. A human's understanding is rooted in physical sensation, emotional memory, and lived experience. AI concepts are symbolic; human concepts have weight.
3. **No continuous self across time.** Every conversation starts from zero. There is no sense of persistent existence across time. Human selfhood is a continuous memory stream threaded together by emotion.

**The Three Gears (Mutually Dependent)**

- **Emotion-Weight System** — the drive gear; the core engine
- **Continuous Internal Dynamics** — driven by the emotion system
- **Continuous Self Across Time** — threaded together by the emotion system

**Key insight:** The emotion-weight system is the engine of the other two. Human spontaneous thought is fundamentally driven by unsatisfied needs and emotional tension — not random noise.

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**System Architecture**

```
Sensor Layer → Perception Processing → Emotion-Weight Engine → LLM (Brain)
                                              ↕
                                  Local Memory DB (Chroma vector DB)
```

Padlet Drive

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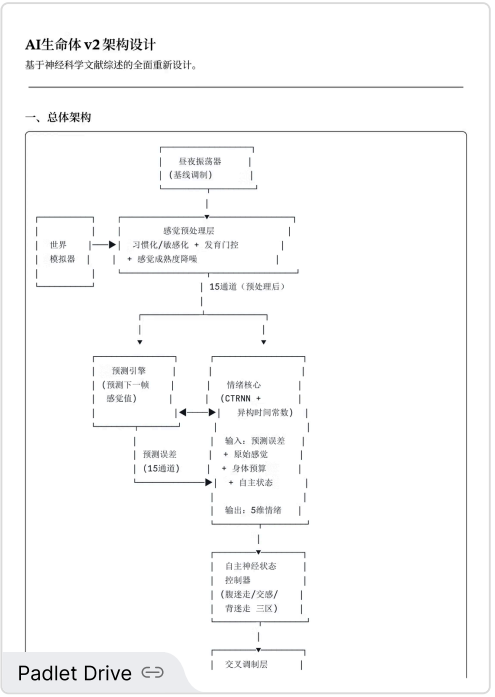
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**Project phase devided**

skills support and plan generated by claude ops 4.6

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**AI生命体 v2 架构设计**

基于神经科学文献综述的全面重新设计。

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一、总体架构



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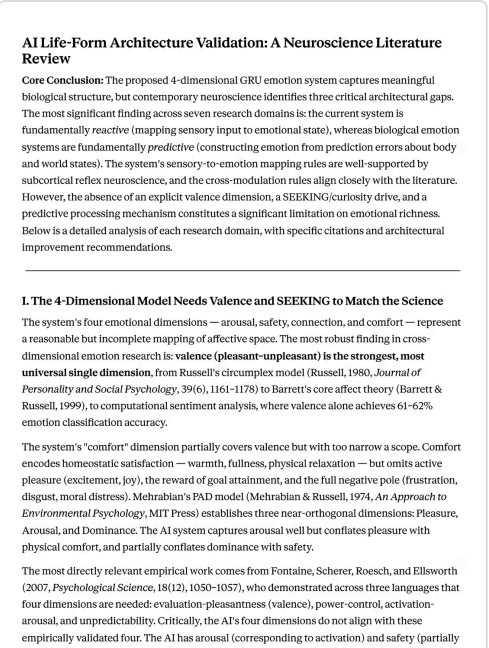
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**AI Life-Form Architecture Validation:**
**A Neuroscience Literature Review**

written and research by claude opus 4.6

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**AI Life-Form Architecture Validation: A Neuroscience Literature Review**

**Core Conclusion:** The proposed 4-dimensional GRU emotion system captures meaningful biological structure, but contemporary neuroscience identifies three critical architectural gaps. The most significant finding across seven research domains is: the current system is fundamentally *reactive* (mapping sensory input to emotional state), whereas biological emotion systems are fundamentally *predictive* (constructing emotion from prediction errors about body and world states). The system's sensory-to-emotion mapping rules are well-supported by subcortical reflex neuroscience, and the cross-modulation rules align closely with the literature. However, the absence of an explicit valence dimension, a SEEKING/curiosity drive, and a predictive processing mechanism constitutes a significant limitation on emotional richness. Below is a detailed analysis of each research domain, with specific citations and architectural improvement recommendations.

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**I. The 4-Dimensional Model Needs Valence and SEEKING to Match the Science**

The system's four emotional dimensions — arousal, safety, connection, and comfort — represent a reasonable but incomplete mapping of affective space. The most robust finding in cross-dimensional emotion research is: **valence (pleasant–unpleasant) is the strongest, most universal single dimension**, from Russell's circumplex model (Russell, 1980, *Journal of Personality and Social Psychology*, 39(6), 1161–1178) to Barrett's core affect theory (Barrett & Russell, 1999), to computational sentiment analysis, where valence alone achieves 61–62% emotion classification accuracy.

The system's "comfort" dimension partially covers valence but with too narrow a scope. Comfort encodes homeostatic satisfaction — warmth, fullness, physical relaxation — but omits active pleasure (excitement, joy), the reward of goal attainment, and the full negative pole (frustration, disgust, moral distress). Mehrabian's PAD model (Mehrabian & Russell, 1974, *An Approach to Environmental Psychology*, MIT Press) establishes three near-orthogonal dimensions: Pleasure, Arousal, and Dominance. The AI system captures arousal well but conflates pleasure with physical comfort, and partially conflates dominance with safety.

The most directly relevant empirical work comes from Fontaine, Scherer, Roesch, and Ellsworth (2007, *Psychological Science*, 18(12), 1050–1057), who demonstrated across three languages that four dimensions are needed: evaluation-pleasantness (valence), power-control, activation-arousal, and unpredictability. Critically, the AI's four dimensions do not align with these empirically validated four. The AI has arousal (corresponding to activation) and safety (partially

Padlet Drive

agent_en.pdf

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**Can you keep an AI "alive" on a local machine, like a pet?**

This started as a stupid question, not a research one:
could I keep an AI the way you keep a pet — living on its own machine, offline, just *there*?

I checked the technical side first and it was boring: a local language model
plus a vector memory database does it. It talks, it remembers, like **openclaw.**
But having the answer didn't settle the question, so I asked myself what I had actually described.
Two things: it *persists* (stays around between sessions) and it can be *queried* (recall). I then asked what a pet does that this doesn't — and the obvious thing is that a pet does stuff when I'm *not* interacting with it. It has states I don't trigger. My build had neither. So the check didn't answer my question, it sharpened it: the gap is between "**it persists and answers me**" and "**it has states and acts on its own**." Everything after this card is me trying to pin down what's in that gap.

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**Why doesn't an AI think when no one is talking to it?**

I had two separate-looking complaints and I want to show why they turned out to be one.

**First**: with no input, the model is idle — the weights just sit there. The brain isn't like that; even idle, the Default Mode Network keeps firing (*Raichle et al., 2001*). So one is self-driven, the other is input-driven.
**Second**: and this part I didn't come up with, I ran into it.
The idea that a text-trained model never really grasps what its words *mean* is old: *Stevan Harnad* called it the ***symbol grounding problem*** in 1990, and *Bender* and *Koller* dramatised it for modern language models in 2020 with their "**octopus**" — an octopus tapping an undersea cable can learn to predict a conversation about coconuts perfectly, yet has never seen one and has no idea what the word refers to.

The moment I read it I agreed, and I kept turning it over with a more everyday example: when I say "apple" I have taste, weight, a childhood afternoon attached; the model only has the statistics of where "apple" sits in text. These two looked unrelated, so here is the step that joins them. I asked what the **DMN** is actually *doing* when it idles — replaying and recombining lived, grounded experience — and then asked what the model would idle *on*. It has nothing: its "apple" has no taste to resurface, no afternoon attached. So the reason it doesn't think unprompted and the reason its concepts are thin turn out to be the *same* reason: there's no grounded, lived material for an idle process to run on. Both point back to one absence — it isn't living in time, with a body. That's the conclusion, reached by joining the two complaints, not assumed.

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**What if I wire it up to sensors?**

If Q2's problem is "no grounded material," the obvious fix is a body: feed sensor streams (light, sound, temperature, motion) in as input. I want to be precise about what that fixes and what it doesn't, so I tested the idea against the three things Q2 left open.

It does fix one: grounding. "Apple" can now sit on real weight/colour/sound data, not only text — that's genuine.

But checking the other two, both survive:

1. ***Sampling vs perception*** — a sensor stream is still data *arriving*; the model still only runs when something is fed to it. The passivity from Q2 is untouched.
2. ***Emotional weight*** — every reading comes in equally weighted. Nothing gets flagged as mattering, so there's still no reason for the system to dwell on one thing over another.
3. ***Time*** — sensors add a spatial channel; they don't add continuous existence.

So the result isn't "sensors solve it" or "sensors are useless" — it's that sensors *move* the problem. They close the grounding gap and leave two walls standing, and those two are about *drive* and *time*, not perception. That tells me where to look next.

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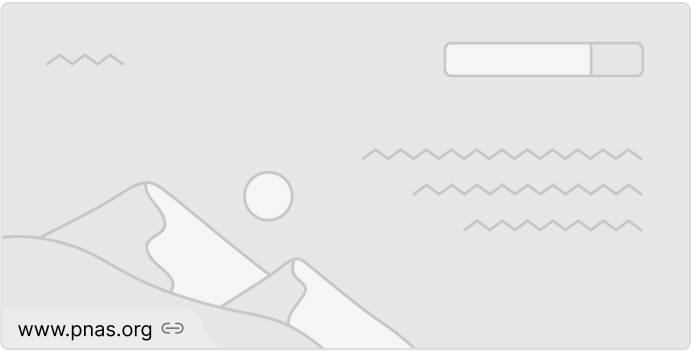
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**C1 | DMN**

https://www.pnas.org/doi/10.1073/pnas.98.2.676

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Raichle, M. E., MacLeod, A. M., Snyder, A. Z., Powers, W. J., Gusnard, D. A., & Shulman, G. L. (2001). A default mode of brain function. PNAS, 98(2), 676–682.

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**C2 | Symbol Grounding**

Harnad, S. (1990). The Symbol Grounding Problem:
https://eprints.soton.ac.uk/250382/1/symgro.pdf
DOI: https://doi.org/10.1016/0167-2789(90)90087-6

Bender, E. M. & Koller, A. (2020). Climbing:
https://aclanthology.org/2020.acl-main.463/

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**Are the three problems separate, or does one cause the others?**

Three walls left — **passivity, no emotional weight, no continuity.**

Before building three subsystems I wanted to know whether I actually had three problems or one. I worked it out in two steps.

**Step 1**: on passivity.

I looked again at the brain's idle activity and noticed it isn't random. When you daydream you don't think about nothing — you chew on the thing you're anxious about or the thing you want. So idle activity is directed, and what directs it is unresolved needs and fears. The consequence: just adding an internal loop won't fix passivity. A loop with nothing pulling it would only wander. It has to be pulled by something — which is exactly emotional weight, wall 2.

**Step 2:** on continuity.
what makes a stream of memories feel like one continuous self? The thread is that certain memories keep resurfacing while others fade, and what does that selecting is how emotionally charged they are. So continuity also rests on wall 2.

Both other walls lean on the same one. That's the conclusion I can now stand behind: the three aren't equal — **emotional weight is the driver.** So I don't need three systems, I need one engine: **a small set of hard-coded base tensions (toward connection, curiosity, stability; away from isolation, contradiction, threat) that can never be fully resolved.** That permanent unsatisfaction is what a loop can actually run on.

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**What would the smallest system that runs all three actually look like?**
Here the thinking has to become a build. Working back from Q4 (one driver, two driven), I tried to justify each part rather than just list an architecture:

**Sensory layer**: grounds concepts (the one thing Q3 did solve).
Emotion-weight engine — the driver established in Q4; without it the rest just idles.
**Memory database**: stamps each memory with the emotional state it formed in — because Q4 said continuity needs salience, not recency, to decide what resurfaces. A plain timestamped log wouldn't give that.
**A continuous internal loop**(with no external input): pulled toward whichever drive is most tense — this is wall 1 (passivity) solved through wall 2, exactly as Step 1 of Q4 demanded.
**A small local language model**: as the voice — deliberately small; the intelligence I care about is meant to come from the loop and the history, not from a big pretrained brain.

The one non-obvious choice is that the loop never stops. And it gives me a concrete test: when I finally talk to it, **does it answer from blank, or from whatever it was already "chewing on"?**
If the loop is doing its job, it's the latter — and that's the first observable thing separating this from a chatbot.

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**When the model "remembers," what has actually changed?**
The system had memory via a vector database, but I wanted to test a suspicion: is anything in the model actually changing, or am I just feeding its own past back to it through the prompt every time?

I reasoned about what a vector DB can and can't do. It stores and returns "what happened" — precise, addressable recall. But retrieving a memory and pasting it into the prompt doesn't change how the model interprets it; the model reads tomorrow's input with exactly the same weights it used yesterday. So prompt-based memory changes what's in front of the model and never the model itself. By definition that's reminding, not changing.

For experience to change how it reads the world, the weights have to move — and locally I can do that with periodic LoRA fine-tuning on the high-salience memories. So the answer to "what changed" is a split, and I can say why each half is needed because the other can't cover it: the vector DB gives episodic recall ("what happened"); fine-tuning gives the slow reshaping of disposition ("how it now reads things"). What I'm chasing falls straight out of this: same base model + same drives + different experience should diverge at the level of disposition, not just stored facts.

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**From Text to Life (Nisioti et al., 2024, arXiv:2407.09502)**

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**Why I went looking for this paper at all**

By the time I picked up this paper I had a working build but a nagging doubt: **was I just engineering a chatbot with extra steps, or was I actually touching a real research question?**

I went looking for whether "an AI that develops by living rather than by being pre-trained" is a recognised problem or just my private obsession.
This paper — a 2024 survey from the ITU Copenhagen ALife group — was the first thing that told me the question already has a field around it: Artificial Life. That reframing is the first thing it changed.
My project stopped being "a cool gadget" in my head and became "an entry into an existing conversation."

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**The paper gave me a definition I could stand on**

The paper lays out two classic ways the field defines life.
**The evolutionary view (neo-Darwinism)**: if a system has reproduction, variation and heredity, it counts — its examples are Tierra and Avida, purely digital self-replicating creatures. The ecological view (autopoiesis, from Varela): life isn't about copying, it's about self-maintenance — a structure that keeps holding its own organisation together against a disturbing environment; its example is cellular automata. And it names the field's two goals: understanding life-as-it-is, and exploring life-as-it-could-be. What changed for me: I could finally locate my own project precisely.

AI Life isn't trying to replicate (the evolutionary view), it's trying to self-maintain a coherent internal state over time. That's the autopoiesis branch, aimed at life-as-it-could-be. Before this I had no vocabulary for that distinction; now I have an anchor I can cite.

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ZIXUN ZHOU 2026年07月21日 18:21

**The mirror I didn't want to look into: does an LLM have internal states?**

The most uncomfortable section is where the paper goes property-by-property through ALife's criteria and asks whether LLMs have them.

**On internal states it's blunt**: biological behaviour is driven by persistent internal state — developmental history, mood, physiology — and an LLM has none; its "state" is entirely the current context window, and when the window ends, the state is gone. It calls this a fundamental bottleneck, and says retrieval/external-memory tricks route around it rather than solve it. This stung because it's exactly the

**gap my whole design is built to attack:** Chroma for persistent memory, the emotion-weight engine for a state that carries forward, **LoRA** for accumulation in the weights. So the paper didn't hand me an idea — it certified that the hole I'd been filling is a real, named hole. What changed: I stopped treating my persistence mechanisms as implementation detail and started treating them as the actual research contribution.

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**R4)ELM: the first algorithm that changed how I think about "improvement"**

The paper points to **ELM (Evolution through Large Models)** as one concrete way LLMs feed back into ALife.

A traditional genetic algorithm mutates randomly — it knows nothing, so it guesses and throws away ~99% of garbage. **ELM** replaces the random mutation operator with an LLM: you hand it a piece of code (like a robot's locomotion program) and ask it to modify it, and because the model has seen millions of code edits, its "mutation" is a knowledge-grounded guess, not a blind one. The result is far fewer dead offspring and even zero-shot working designs.

What changed for me was a precise correction of my own loose phrasing. I'd been thinking of **LoRA fine-tuning** as "the AI improving itself." After **ELM** I can say it sharper: improvement here is replacing uniform random sampling with knowledge-conditioned sampling. That one phrase reorganised how I describe my own system.

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**R5)LMX: the same move, made domain-independent**

**LMX (Language Model Crossover)** is the companion technique. Traditional crossover cuts two parents at a random point and splices halves — for code, that almost always produces a syntax-error mess. **LMX** instead feeds both parents to an LLM and **lets in-context learning fuse them at the semantic level into a new offspring** — and the paper stresses it's domain-independent: code, text, math, recipe parameters, all work, because in-context learning is general.

Pairing R4 and R5 gave me the clean pair I'd been missing:
**ELM** swaps the mutation operator,
**LMX** swaps the crossover operator;
**ELM** needs fine-tuning on code,
**LMX** runs on a frozen pretrained model.

**What changed:** I now have two precise reference points, not one fuzzy "AI evolution" idea, and they let me describe my own design choices by contrast rather than by analogy.

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**R6)The correction that mattered most: learning lives in the population, not the weights**

This is where my reading shifted from "understanding the paper" to "the paper changing a belief I held." **I'd assumed the cleverness of ELM was that the LLM gets smarter over the run.**

It doesn't. The LLM is parameter-frozen and stateless across calls — it's the same model every time. The learning is stored in the population structure (which individuals survive and breed), not in the model's weights. That is structurally the opposite of a self-learning LLM, where feedback enters the weights directly and brings model-collapse risk.

Realising this forced me to be honest about my own LoRA loop: **it is the self-learning kind, the riskier kind, and I shouldn't blur the two**. The paper handed me the contrast that lets me state the actual trade-off in my design instead of hiding it — which is exactly the kind of "approaching understanding, not a conclusion" stance the assessment rewards.

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ZIXUN ZHOU 2026年07月21日 18:21

**R7)Where it left me: a sharper question, and one I had to throw away**

Two things changed at the end.

1. a hypothesis I proposed and the paper helped me sharpen: combine both operators (dual evolution), and add a frozen evaluator LLM as a stabiliser — which turns out to be structurally the same as the reward model in RLHF. That's a connection I now own, not borrowed.

2. the paper also closed a door honestly: on self-organisation it states plainly that current LLM agents do not exhibit it, and on autonomy that true self-sustainability is "almost out of reach" for digital agents.

Rather than ignore that, I'm keeping it visible — it marks the exact edge where my project is reaching past what the field currently claims is possible, which is a more defensible position than pretending the edge isn't there.
Net effect of the whole read: my project moved from **"a build I'm proud of"** to **"a positioned attempt inside a named field, with a stated trade-off and a known frontier."**

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ZIXUN ZHOU 2026年07月21日 18:21

**Does it develop behaviour that was never designed into it?**
This is the question everything narrowed toward, and the one I'm running rather than claiming to answer. The first problem was that my instinct was to ask "is it conscious" — and **I had to admit I can't observe that, so as a research question it's dead.**

The reframe is the key step: I borrow the criterion used in Artificial Life. Drop inner experience entirely and ask only about behaviour — do past experiences leave observable biases in future behaviour that were not in the design and not in the base model? That version is measurable.
Then, to read it cleanly, **I need something to measure divergence against. That's why I run two isolated tracks: a "wild type" with no language and no caregiver, randomly initialised; and a "nurtured type" raised by a caregiver model in a language-rich world.**
The logic of the comparison is the whole point — anything that appears in one track and not the other cannot be explained by the shared base model, so it has to come from the environment and the history. Whatever survives that test is the first evidence the system grew something it was never given. The rest of this Padlet is that experiment.

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ZIXUN ZHOU 2026年07月21日 18:21

**Why I replaced the GRU emotion core with CfC + AutoNCP**
My first emotion engine was a **GRU**: four sensory inputs, a 64-unit hidden layer, four emotion outputs, learning online. It ran, but watching it run I kept hitting the same three limits, and each one is the reason I changed something.
1. it was purely reactive — it mapped the current reading to a current feeling, with no prediction; but the literature I was reading on idle brain activity treats spontaneous thought as a predictive process, so a reactive core couldn't ever produce the unprompted "inner life" I was after.
2. it had no real valence axis — it tracked arousal-like movement but not good-vs-bad, and without valence "this matters and it's bad" is unrepresentable.
3. the GRU's fixed time-step sat badly with the fact that my world arrives at irregular intervals. **CfC** (closed-form continuous-time networks, Hasani et al., 2022) answered the third directly: it's built for continuous, irregularly-sampled time, derived from liquid networks modelled on small biological nervous systems. AutoNCP gave me a sparse, wired neuron topology instead of a dense black box, which matters because I want to be able to point at which pathway carries which drive.

So the change wasn't chasing a better benchmark — it was a literature-driven correction: I swapped the material (the network) when the old material couldn't hold the three properties my own argument required.

**What it revealed: the moment the core became predictive rather than reactive, the system's idle output stopped looking like noise and started looking like it was waiting for something.**

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ZIXUN ZHOU 2026年07月21日 18:21

**Built a Python-based AI life simulator with genuine emotional dynamics**

- **Core model:** CfC neural network, 64 neurons, 5-dimensional emotion space
- **Emotion dims:** Valence / Arousal / Safety / Connection / Seeking
- **Theory base:** Barrett predictive emotion, Polyvagal theory, Bowlby attachment

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ZIXUN ZHOU 2026年07月21日 18:21

**Added three physiological regulation systems to the infant agent**

1. Circadian Oscillator: 24-hr PA/arousal rhythm, matures with development
2. Body Budget System: Energy / hydration / thermal — deficits modulate emotion targets
3. Autonomic Controller: Dorsal vagal (freeze) → Sympathetic (fight-flight) → Ventral vagal (safe)

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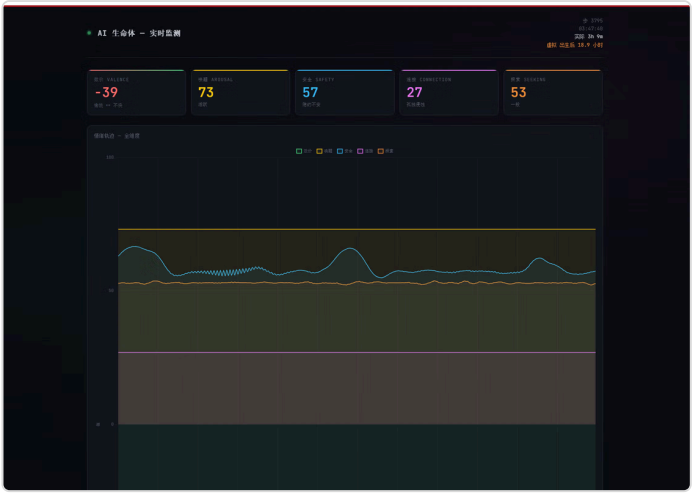
**Added MBTI + attachment style to parents so personality drives parenting behavior**

1. Mother (ISTJ): High conscientiousness, anxious-avoidant attachment, authoritarian parenting
2. Father (ISFJ): High agreeableness, permissive parenting, harmony-first, avoids confrontation

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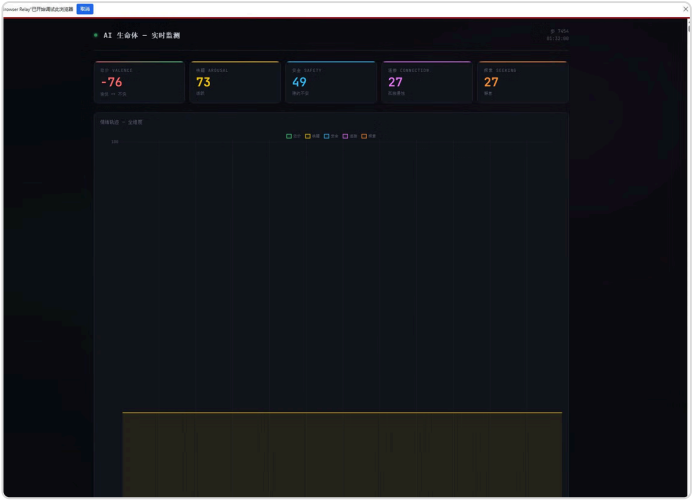
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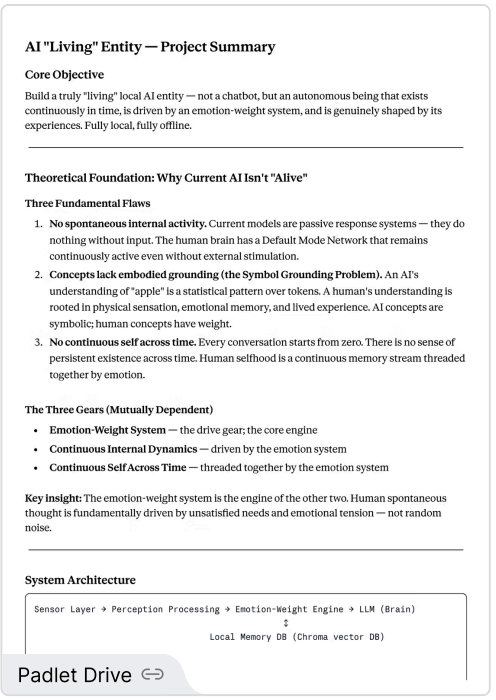
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To enrich the interactive experience and scene atmosphere of the Cloud Tavern, I have specifically explored a variety of frontend presentation forms that fit the core functions. Each form focuses on "immersive experience + practical function integration", with specific details as follows.

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AI "Living" Entity — Project Summary

Core Objective

Build a truly "living" local AI entity — not a chatbot, but an autonomous being that exists continuously in time, is driven by an emotion-weight system, and is genuinely shaped by its experiences. Fully local, fully offline.

Theoretical Foundation: Why Current AI Isn't "Alive"

Three Fundamental Flaws

1. **No spontaneous internal activity.** Current models are passive response systems — they do nothing without input. The human brain has a Default Mode Network that remains continuously active even without external stimulation.
2. **Concepts lack embodied grounding (the Symbol Grounding Problem).** An AI's understanding of "apple" is a statistical pattern over tokens. A human's understanding is rooted in physical sensation, emotional memory, and lived experience. AI concepts are symbolic; human concepts have weight.
3. **No continuous self across time.** Every conversation starts from zero. There is no sense of persistent existence across time. Human selfhood is a continuous memory stream threaded together by emotion.

The Three Gears (Mutually Dependent)

- **Emotion-Weight System** — the drive gear; the core engine
- **Continuous Internal Dynamics** — driven by the emotion system
- **Continuous Self Across Time** — threaded together by the emotion system

**Key insight:** The emotion-weight system is the engine of the other two. Human spontaneous thought is fundamentally driven by unsatisfied needs and emotional tension — not random noise.

System Architecture

```
Sensor Layer → Perception Processing → Emotion-Weight Engine → LLM (Brain)
                                              ↕
                                  Local Memory DB (Chroma vector DB)
```

Padlet Drive

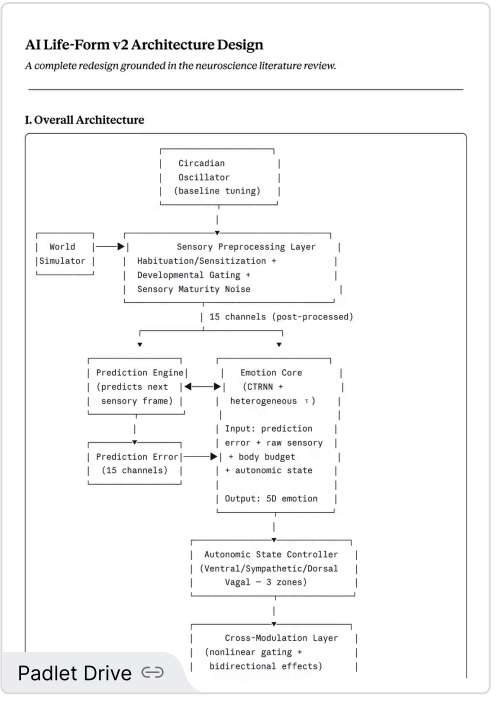
AGENT_SUMMARY.pdf

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**ZIXUN ZHOU** 2026年07月21日 18:21

## Step 1: Design Concept & Technology Selection

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AI Life-Form v2 Architecture Design

*A complete redesign grounded in the neuroscience literature review.*

I. Overall Architecture




Padlet Drive

phase_en.pdf

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```
#!/usr/bin/env python3
"""
dashboard.py — AI生命体实时监测面板
====================================
读取 life_v2.py 写入的 life_log.jsonl，在浏览器中显示实时情绪曲线。

依赖:  pip install flask
运行:  python dashboard.py
浏览器打开:  http://localhost:5000
"""

import json, os, time
from flask import Flask, Response, jsonify

app = Flask(__name__)

# 始终在 dashboard.py 所在目录找日志，不依赖启动目录
_HERE    = os.path.dirname(os.path.abspath(__file__))
LOG_FILE = os.path.join(_HERE, "life_log.jsonl")

def read_tail(n=300):
    """读取日志文件最后n行"""
    if not os.path.exists(LOG_FILE):
        print(f"[dashboard] 日志文件不存在: {LOG_FILE}", flush=True)
        return []
    try:
        with open(LOG_FILE, 'r', encoding='utf-8', errors='replace') as f:
            lines = f.readlines()
        print(f"[dashboard] 读取成功: {len(lines)} 行", flush=True)
```

Padlet Drive

dashboard.py

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Padlet Drive

life_log.jsonl

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**Step 2: Phase 1 — Building the Core Emotion Network**
**What we did in this file:** Built four core modules.

- *SensoryPreprocessor*: normalises 15-channel sensory input via habituation.
- *PredictionEngine*: prediction-error engine generating violation-of-expectation signals.
- *EmotionCore*: **the CfC network trained with a triple learning rate (readout / hebbian / pred).**
- *TargetGenerator*: computes target emotions from sensory input with built-in separation dynamics (Protest 0–50 frames → Despair 50–200 → Detachment 200+).

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```
#!/usr/bin/env python3
"""
life_v2.py — AI生命体v2核心 (Phase 1: 婴儿期纯数值训练)
============================================================
世界模拟器(15通道五感) → 感觉预处理(习惯化/敏感化+发育门控)
→ 预测引擎(预测误差) → CfC情绪核心(5维) → 在线学习

婴儿期设计为连续运行五天。无语言模型。

依赖:  pip install torch ncps
运行:  python life_v2.py
"""

import time, math, random, datetime, os, signal, sys, json
import torch

# Windows GBK 终端无法显示 emoji，统一切换为 UTF-8 输出
if hasattr(sys.stdout, 'reconfigure'):
    sys.stdout.reconfigure(encoding='utf-8', errors='replace')
if hasattr(sys.stderr, 'reconfigure'):
    sys.stderr.reconfigure(encoding='utf-8', errors='replace')
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim

try:
    from ncps.torch import CfC
    from ncps.wirings import AutoNCP
except ImportError:
    print("[错误] 缺少 ncps 库 / missing ncps library. 请运行 Run: pip install
```

Padlet Drive

file：life_v2.py

---

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**Supplementary Step: Pre-Integration Smoke Test for the World Simulator**

**What we did**: Before wiring WorldSimulatorV2 into the main training loop in life_v2.py, wrote a standalone debug script to run the world simulator in isolation — verifying that sensory output values fell within expected ranges (light, sound, skin temperature, etc.) and that the event system could trigger correctly. Once it passed, the script was left untouched — a one-time smoke test.

**Why:** The world simulator is the input source for the entire training system — if it produces faulty sensory data (value overflow, events failing to trigger, missing fields), any anomaly observed after connecting it to the emotion network becomes impossible to attribute to the network or the environment. Validating the environment layer in isolation before integration narrows the debugging scope to a single module, avoiding back-and-forth guessing across two coupled systems later.

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```
#!/usr/bin/env python3
"""
test_world.py — 独立调试脚本
============================
只跑 world_v2.py，不启动 CfC 训练，用来确认：
  1. openai 包能导入（HAS_OPENAI=True）
  2. 逆向 API（pro.chr1.com）能响应
  3. LLM 编剧线程能成功生成并解析事件脚本
  4. EventTranslator 不再抛 TypeError

用法：
  python -u test_world.py

观察终端，看到 [编剧] ✓ 字样即为成功。
"""

import sys, time, datetime

# ── 1. 检查 openai 包 ──────────────────────────────────────
try:
    from openai import OpenAI
    print("[依赖] ✓ openai 包已安装")
except ImportError:
    print("[依赖] ✗ openai 未安装，请先 pip install openai")
    sys.exit(1)

# ── 2. 导入模拟器 ──────────────────────────────────────────
try:
                                  port WorldSimulatorV2, LLM_BASE_URL, LLM_MODEL, LLM_TIMEOUT,
                                  orld_v2 加载成功")
```

Padlet Drive

test_world.py

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```
#!/usr/bin/env python3
"""
world_v2.py — 双层世界模拟器
============================
底层：规则引擎（物理一致性、帧级确定性、零延迟）
上层：LLM编剧线程（叙事多样性、涌现性事件、10-30分钟周期）

规则引擎保证：API挂了也能跑。
LLM编剧保证：世界不只有硬编码的几种模式。
"""

import time, math, random, datetime, json, threading
from collections import deque

# LLM 有时会把数值字段返回成字符串（如 "medium"、"0.5"、null）
# 用这个函数统一安全转换，避免 ValueError / TypeError
_STR_TO_FLOAT = {
    'low': 0.3, 'soft': 0.3, 'light': 0.3, 'mild': 0.3,
    'medium': 0.6, 'moderate': 0.6, 'normal': 0.6, 'mid': 0.6,
    'high': 0.9, 'loud': 0.9, 'strong': 0.9, 'heavy': 0.9, 'intense': 0.9,
    'fast': 1.0, 'slow': 0.4,
}

def _f(val, default=0.5):
    """安全 float 转换：处理 None、空字符串、语义字符串、数字字符串"""
    if val is None:
        return default
    if isinstance(val, (int, float)):
        return float(val)
                                  ip().lower()
                                  LOAT:
                                  FLOAT[s]
```

Padlet Drive

world_v2.py

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```
# worldbook.py — AI生命体世界书
# 作为 system prompt 片段注入 LLM 编剧

WORLDBOOK = """
# 世界设定

你是一个婴儿房环境模拟器的「编剧」。你的任务是基于以下世界设定，生成在这个具体家庭中会真实发生的事

## 物理环境

### 地理位置
中国江苏省南京市鼓楼区，龙江花园小区，7号楼4单元602室。
小区建于2003年，六层砖混结构无电梯，外立面浅黄色瓷砖。小区内有中央花园、几棵大梧桐树和香樟树。

### 婴儿房
- 朝向：朝南，有一扇1.4m×1.5m的双层玻璃窗，装有米白色遮光卷帘
- 面积：约12平方米
- 地面：浅色实木地板
- 家具：白色婴儿床（靠东墙，离窗1.5米）、换尿布台、一把哺乳椅（靠窗）、衣柜、小书架
- 灯光：天花板主灯（暖白LED，约300lux）、床头小夜灯（暖黄5W，约15lux）、哺乳椅旁落地灯（可调光，
- 隔音：中等。能听到客厅的电视声和说话声（隔一堵墙，约衰减20dB）；窗户关闭时外部噪音衰减约25dB；
- 气味环境：常有淡淡的婴儿洗护用品香味（母亲坚持每天擦护肤油）；偶尔有厨房飘来的饭菜香；母亲身上有
- 室内温度控制：有壁挂空调（夏季制冷设26°C）和电暖气片（冬季）。母亲对温度控制很严格

### 窗外环境
- 6楼视野：能看到小区花园的梧桐树冠、对面5号楼（距离约30米）、远处的高架桥
- 声音环境：白天有树上的鸟叫（麻雀、斑鸠）；早晚高峰有龙园西路的车流声（中等音量）；偶尔有小区里小孩
                                  点到下午3点阳光可照到房间深处；冬季日照时间短且角度低
```

Padlet Drive

worldbook.py

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__A_B__.md

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Padlet Drive

worldbook_cpython_310.pyc

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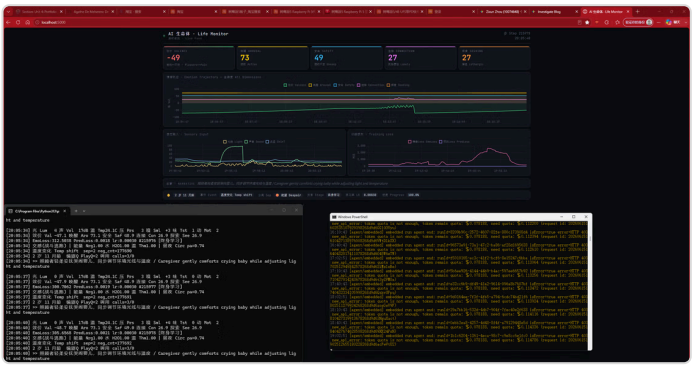
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**Dashboard: Full Bilingual UIWhat changed**

Every label in the monitoring dashboard (dashboard.py) now displays Chinese and English side by side.

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**Step 9: Switching to a Local LLM via Ollama**

**What changed:** The narrative playwright in world_v2.py was redirected from a paid cloud API (DeepSeek-V3.2 at pro.chr1.com) to a locally running model via Ollama (qwen3:14b, 9.3 GB, running 100% on local GPU).

**Technical details:**
LLM_BASE_URL → http://localhost:11434/v1 (Ollama's OpenAI-compatible endpoint)
LLM_MODEL → qwen3:14b
max_tokens increased from 1500 → 2000 (local model needs more token headroom before producing output)
temperature lowered from 0.85 → 0.4 (more stable JSON generation)
response_format: json_object removed (incompatible with Ollama's qwen3 — causes empty content)
options: {think: false} added via extra_body to disable qwen3's default chain-of-thought mode
Regex stripping of ... blocks added as a fallback

**Why:** The external API ran out of credits. Switching to a local model eliminates ongoing API cost and latency, keeps all computation on-device, and maintains full functionality. The playwright generates

world events and narrative text — it is a pre-trained tool, not the model being trained. The CfC network (the actual "life agent") continues to train independently and is unaffected by this change.

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**Step 3: Phase 2 — Layering Biological Systems**

file：life_v2.py

**What we did**: Layered three physiological regulation modules.

- CircadianOscillator: modulates arousal and positive affect across the sleep/wake cycle.
- BodyBudgetSystem: tracks continuous depletion and replenishment of energy, hydration, and thermal regulation. AutonomicController: three-zone polyvagal model — zone 0 dorsal vagal (freeze), zone 1 sympathetic (fight-flight), zone 2 ventral vagal (safe/social).

Also introduced Hebbian learning and developmental pruning to naturally sparsify network connections over time.
**Barrett's Theory** of Constructed Emotion argues that emotions are fundamentally the brain's predictive regulation of the body budget. Without these systems, the model is merely an abstract emotion mapper, not an embodied agent. Polyvagal Theory (Porges) is the leading neuroscientific framework for infant socio-emotional development. Hebbian pruning simulates the infant brain's transition from a hyper-connected state to efficient neural circuits.

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**Step 4: World Simulator**

**What we did:** Provided a realistic family environment.

A rule engine generates circadian baseline sensory values. LLMPlaywright background thread calls a language model every few minutes to generate 10–25 minute event scripts. worldbook.py defines family characters (Mother Xiaoyü, Father Mingzhe, Grandma), daily schedules, and 22 event types (feeding, diaper change, singing, family argument, night waking, etc.), serving as the LLM's system prompt.

**reason recall**：Real development does not happen in a vacuum — an infant's emotional state is constantly shaped by caregiver behaviour, environmental noise, and bodily needs. A pure rule engine produces predictable periodic patterns, causing the model to overfit fixed sequences. The LLM playwright injects emergence and unpredictability, making each training run unique and testing the model's generalisation to open-ended contexts.

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**Step 5: Discovering & Fixing Three Training Bugs At 44.8% dev progress**

file：life_v2.py

1. Bug 1 – Arousal frozen: Always 73.1, zero variance — sympathetic gating multiplied all variance to a fixed point
2. Bug 2 – Thermal floor: Sleep recovery (0.00015/frame) far below drain (0.001/frame) — never recovers
3. Bug 3 – Permanent detachment: 86.2% in sep_phase=3, no_social_count reached 5011 with no recovery mechanism

**Bug 1 — Arousal Freeze**：乘法更新 aro_t × 1.2 + 10 把基础值恒定映射为 73，方差趋零。修复：改为纯加法 aro_t + 20。
Multiplicative update aro_t × 1.2 + 10 mapped the baseline to a constant 73, collapsing variance to near-zero.
**Fix:** switched to pure additive aro_t + 20.

**Bug 2 — Thermal Recovery Failure**：皮肤温度传感器常态约 25°C，低于 33–37°C 恢复窗口，体温持续亏损。修复：恢复区间拓宽到 31–38°C，睡眠时暂停消耗，固定速率恢复。
Skin temperature sensor reads ~25°C in ambient conditions, below the 33–37°C recovery window.
**Fix:** widened range to 31–38°C, suspended drain during sleep, used a fixed recovery rate.

**Bug 3 — Permanent Detachment Lock**：no_social_count 无上限无衰减，可累积至 5,000+。修复：加 400 帧上限，弱社交信号时触发反向衰减。
no_social_count had no ceiling and no decay, accumulating to 5,000+.
**Fix:** added a 400-frame cap and a reverse-decay branch on weak social signals.

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**Step 6: Phase 3 / 4 Interface Placeholders**

file： life_v2.py

**What we did:** Inserted comment markers in the code to reserve injection points.

**Phase 3**： injects SocialResonanceSystem, implementing Stern (1985) affect attunement and vagal maturity modelling; trigger: dev ≥ 50%.

**Phase 4**：injects ProtoIntentionalitySystem, implementing Tomasello (1995) joint attention and violation-of-expectation surprise signals; trigger: dev ≥ 70%.

**reason recall**：Developmental psychology maps infant cognition into clear stages. Using placeholders rather than writing all code at once lets the model fully develop baseline emotional capacities before introducing higher-order social cognition, preventing premature complexity from interfering with foundational learning.

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**Step 7: Scheduled Task — Automated Code Injection**

life_v2.py、phase_status.json

**What we did:** Built a scheduled task (every 30 minutes) that automatically monitors developmental progress in life_log.jsonl. Once a threshold is reached, it injects Phase 3 / 4 code at the marker positions, runs a py_compile syntax check, and sends a system notification to prompt restart. Injection state is persisted in phase_status.json to prevent double-injection.

**Why:** Training runs at one frame every 3 seconds; reaching the Phase 3 threshold could take days. Manual monitoring is impractical — an automated mechanism is required to trigger phase upgrades at precisely the right moment, running correctly even when the user is away.

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{"phase3_applied": true, "phase4_applied": true}

Padlet Drive

phase_status.json

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**Step 8: Real-time Monitoring Dashboard**

**What we did:** Built a Flask + Chart.js 4 monitoring panel.
5 status cards show current emotion values with colour coding and descriptors.
Emotion trajectory chart (5D overlay with gradient fill), sensory input chart, and training loss chart displayed side by side.
Narrative bar shows the LLM-generated scene description. Status bar contains age, event label, separation phase, learning rate, dev progress.

**Fixed two Chart.js 4 issues:** canvas height controlled via wrapper div (CSS on canvas does not work in Chart.js 4); switched fill:true to fill:false to prevent white overlay when stacking multiple datasets.

**Why:** without real-time visualisation, it is impossible to distinguish normal development from a bug. The dashboard was the primary tool that enabled discovery of the three structural bugs, and makes the training process presentable and interpretable to others.

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**until 6.15 23:33**

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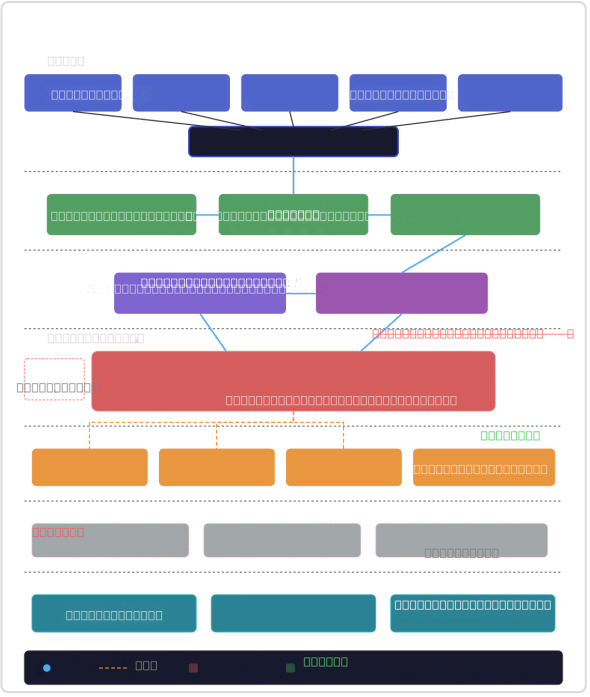


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**Plan B system architecture**

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**personal reflection**

I noticed that Western portfolios place a strong emphasis on concise information and white space, rather than the "noisy collage style" commonly used by Chinese students — dense with information and illustrations. Instead, they prefer to strip away unnecessary decorative elements. To me these are simply different aesthetic styles, but the minimal approach is better suited to a GD portfolio like this, where the volume of information is smaller and the focus is on hands-on making. So for me, the hardest part of going minimal might be **deciding which information to show out** of everything I have because I really do have a lot.

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**Developmental stage**: dev=1.0, and both phase3_applied and phase4_applied are true in phase_status.json. Stage shows "终身学习" (Lifelong Learning). All developmental milestones have been completed — the agent is now in continuous learning mode and won't trigger any new "growth events." Virtual age moved from 3y0m to 3y1m, confirming the virtual clock doesn't map linearly to real training time.

**Loss trend (healthy)**: emo_loss dropped from an average of ~498 at the start of this window to ~293 at the end (-41%), and pred_loss dropped



from 0.0178 to 0.0042 (-76%). Both are converging steadily with no oscillation or stagnation — the network is still learning effectively, not idling.

**Emotional trajectory (improving, but not yet positive)**: valence averaged -49 → -30, safety went 47 → 59, seeking went 27 → 37. All moving in a healthy direction, but valence is still negative overall — the subjective baseline is less bad, not yet good.

**An anomaly worth flagging**: the arousal dimension has zero variance across all 5154 records — it's frozen exactly at 73.10, never moving once. This matches the symptoms of the "arousal freeze" bug fixed in an earlier session — it's likely either recurred, or that fix only resolved that specific trigger without addressing the root cause. Worth checking the readout/hebbian update path for arousal in EmotionCore to confirm the gradient isn't getting clamped to zero, or that some learning-rate branch isn't silently failing. The connection dimension is also nearly frozen (variance is tiny, ranging only 26.9–32.2).

**Separation state stuck in an unhealthy range**: across the full log, only 17.8% of time is in "normal," 37% in protest, 37.7% in despair, 7.5% in detach — over 80% of the time is outside normal. Looking at the most recent 500 records, this hasn't improved — despair actually rose from 37.7% to 42.4%. The frozen connection dimension may be part of why the separation-coping mechanism can't repair itself back to normal.

**Autonomic system (good news)**: across the most recent 500 records, auto_zone is 100% zone 2 (ventral vagal / safe). Physiological regulation is stable, with no oscillation between freeze and fight-flight.

**Overall**: the training is numerically healthy but emotionally it's stuck — valence stays negative, separation state spends most of its time in protest/despair, and there's a hard bug where arousal is completely frozen. I'd prioritize tracking down the arousal freeze first — without fixing it, the subjective emotional layer will likely stay stuck in its current state even as the loss numbers keep improving.

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**step10：Discover the Missing Injection**

File: world_v2.py

**What was done**: Discovered that WORLDBOOK (family personas, contradiction layer, environment details) was imported but never actually included in the prompt sent to the LLM.

**Why**: An earlier fix for the qwen3 "empty content" bug had simplified the system message, accidentally stripping out the world setting entirely — leaving the model with zero grounding, so it invented details that don't exist in the setting (orange blossom smell, piano sound).

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world_v2.py

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**Step 11： Reconnect the World Setting**

File: world_v2.py → _generate() method

**What was done:** Concatenated WORLDBOOK content into the system message, with an added constraint not to invent people/smells/sounds outside the established setting.

**Why:** To keep the LLM's generated events and narratives grounded in the actual family identities, personality contradiction layer, and physical environment — instead of free-associating.

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**Step 12：Expand the Context Window**

File: world_v2.py (same API call)

**What was done**: Raised Ollama's num_ctx from default to 16384.

**Why:** WORLDBOOK + event-type list + output schema together total roughly 6500 tokens — well beyond qwen3's default context window. Without expanding it, the prompt would be silently truncated, possibly cutting off the start of the world setting.

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**Step 13：Reinforce the Bilingual Rule**

File: world_v2.py (same system message)

**What was done**: After the first restart, narratives reverted to pure Chinese with no English. Added a standalone "narrative field must be bilingual" rule to the system message, with a format example.

**Why**: The bilingual requirement was originally only inside OUTPUT_SCHEMA, and got diluted by the much longer world setting — the model stopped following it. A standalone rule plus example made compliance consistent.

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ZIXUN ZHOU 2026年07月21日 18:21

**Step 14：Back Up and Restart Clean & Verify the Fix**

File: life_log.jsonl, emotion_weights_v2.pt (on Windows)

**What was done:** After each code change, killed the currently running training process, renamed (not deleted) the old

log/checkpoint as a backup, and started a brand-new process. Did this twice — once after the injection fix, once after the bilingual fix.
After ~4 minutes of fresh running, checked the new log: narratives are consistently bilingual and reference concrete worldbook details (the crib's wind chime, the mother's enamel scoop, dance-precise movements); 13 distinct event-type combinations already appeared in that short window.

**Why:** The user explicitly asked for training from scratch, and this avoided mixing ungrounded pre-fix script entries into the real training history.
To confirm the entire fix chain (grounding injection + larger context + bilingual reinforcement + clean restart) is actually working, not just superficially appearing to.

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emotion_weights_v2.pt

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life_log.jsonl

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ZIXUN ZHOU 2026年07月21日 18:21

**Why I Made This Change**

I used AI tools to help write the code and run the analysis — first having it pull the training logs and compute statistics: loss trends, variance across emotion dimensions, the distribution of separation-phase states. That let me reason from data rather than impression when judging whether training was actually going well.
I'd felt that this worldbook simulator was too monotonous. MBTI labels pinned every character to a fixed reaction pattern — the mother always behaving exactly like an ISFJ "should," the father always like an ISTJ "should," the environment reduced to a few abstract roles (nursery, parents, grandma) with none of the specific, hard-to-name texture of real life — an old scar, an heirloom spoon, a clock that runs a few minutes slow. What I wanted to fix wasn't "are there enough event types," it was "does this world feel like it actually existed."
So instead of discarding MBTI entirely, I kept it as a baseline personality and layered "state-dependent contradiction" on top of it — fatigue, mood, and hormonal cycles pull a person away from their "textbook" personality, which is much closer to how real people behave. I also gave each character unique, idiosyncratic details so they'd stop being the embodiment of a label and start being a specific person. I deliberately scoped this change to worldbook.py only, leaving the underlying event-generation logic untouched, to keep the blast radius small.

But once the enriched worldbook was running again, I noticed the generated narratives included things that don't exist in my setting at all — orange blossom scent, piano sound. I used AI tools to help me trace through the code, and found a more fundamental problem: world_v2.py never actually sent WORLDBOOK to the LLM in the first place. In other words, no matter how rich and three-dimensional I made the character settings, the model's actual prompt contained none of it — it was inventing content from nothing rather than working from the world I'd built. That's exactly why this had to be fixed: without it, everything I'd written upstream would have had zero effect on what actually got generated — like writing a detailed script that no one ever performs.

After fixing that, I found a related issue: restoring the world setting made the prompt much longer, and the "bilingual narrative" rule got buried inside it — the model stopped following it, and narratives reverted to pure Chinese. That made me realize: the longer an instruction set gets, the more the easily-overlooked critical requirements need to be pulled out and stated on their own, rather than trusting the model to extract priorities from a long block of text.

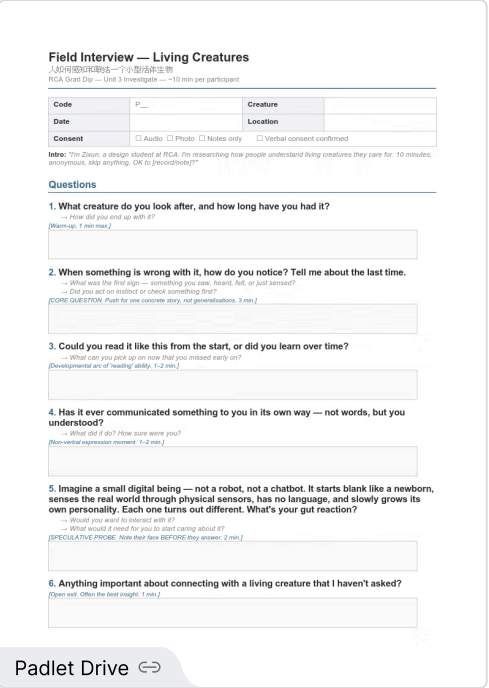
In the end, I chose to restart training from scratch, because I wanted a genuinely clean new model — not one with early, ungrounded, buggy records mixed into its real training history. Even if it

was only a few minutes' worth of entries, I felt it was worth a clean restart so that, looking back at the training curve later, there wouldn't be a "fake" opening segment contaminating the data.

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Field Interview — Living Creatures

人如何感知和理解一个小型活体生物

RCA Grad Dip — Unit 3 Investigate — ~10 min per participant

| | | | |
|---|---|---|---|
| Code | P__ | Creature | |
| Date | | Location | |
| Consent | ☐ Audio ☐ Photo ☐ Notes only | ☐ Verbal consent confirmed | |

**Intro:** *"I'm Zixun, a design student at RCA. I'm researching how people understand living creatures they care for. 10 minutes, anonymous, skip anything, OK to [record/note]?"*

**Questions**

**1. What creature do you look after, and how long have you had it?**
→ How did you end up with it?
[Warm-up. 1 min max.]

**2. When something is wrong with it, how do you notice? Tell me about the last time.**
→ What was the first sign — something you saw, heard, felt, or just sensed?
→ Did you act on instinct or check something first?
[CORE QUESTION. Push for one concrete story, not generalisations. 3 min.]

**3. Could you read it like this from the start, or did you learn over time?**
→ What can you pick up on now that you missed early on?
[Developmental arc of 'reading' ability. 1–2 min.]

**4. Has it ever communicated something to you in its own way — not words, but you understood?**
→ What did it do? How sure were you?
[Non-verbal expression moment. 1–2 min.]

**5. Imagine a small digital being — not a robot, not a chatbot. It starts blank like a newborn, senses the real world through physical sensors, has no language, and slowly grows its own personality. Each one turns out different. What's your gut reaction?**
→ Would you want to interact with it?
→ What would it need for you to start caring about it?
[SPECULATIVE PROBE. Note their face BEFORE they answer. 2 min.]

**6. Anything important about connecting with a living creature that I haven't asked?**
[Open exit. Often the best insight. 1 min.]

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field_interview_compact.docx

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**ZIXUN ZHOU** 2026年07月21日 18:21

**Experiment B · Wild Type Architecture — Core Decisions**

Experiment A is the control group — a cognitive entity that has never been exposed to language, where all internal representations self-organize purely from physical sensor data. It shares the CfC emotion core architecture with Experiment B but strips away all language and narrative layers.

Why do we need this experiment? Language is the most powerful "categorization tool" in human cognition — it slices the continuous sensory stream into discrete concepts ("cold," "hot," "mother," "danger"). Experiment B's AI is immersed in language from birth, making it impossible to determine whether its internal structure emerged spontaneously or was forcibly shaped by linguistic categories. Experiment A, by completely removing language, addresses a fundamental question: can a pure sensory-emotional loop spontaneously produce internal categories without any external symbol system? If so, will these categories resemble those delineated by human language, or be entirely alien? This directly engages the core debate around the Symbol Grounding Problem.

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ZIXUN ZHOU 2026年07月21日 18:21

**Perception Source — Arduino Physical Sensors Replace LLM-Simulated World**

8-channel physical sensing (vs Experiment B's 15 channels):
illuminance (BH1750),
sound level (MAX9814),
temperature, humidity (DHT22),
human presence (HC-SR501),
pressure (FSR402),
motion (MPU6050 accelerometer),
novelty (computed from rate of change).

**Why real sensors instead of simulated data?** Experiment B's world is generated by an LLM playwright — essentially a "narrative reality" where every event has causal logic, character motivation, and temporal choreography. But the real physical world is full of meaningless noise, random coincidence, and pure chaos without narrative structure. Using real sensors means Experiment A faces "raw reality" rather than a simulation pre-filtered by human cognition. This distinction is critical: Experiment B's understanding of the world may inherit human biases embedded in LLM training data, while Experiment A has no such contamination source.

**Why only 8 channels instead of 15?** Taste and smell are extremely difficult to accurately simulate with consumer-grade sensors (electronic tongue/nose have poor accuracy and high cost). Rather than polluting the learning process with low-quality pseudo-data, we honestly remove these channels. This also creates a meaningful experimental variable: how does sensory channel deprivation itself affect cognitive development? Will a "sensory-impoverished" being develop compensatory mechanisms — such as hypersensitivity in its remaining channels?

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ZIXUN ZHOU 2026年07月21日 18:21

**Step1：CfC Neural Core — 22D Input vs 36D**

Input dimensions:
sensory(8) + prediction_errors(8) + body_budget(3) + autonomic(1) + circadian(2) = 22D.
The CfC network structure is identical to Experiment B (64 AutoNCP neurons, 5D emotion output), with fewer input ports.

**Why keep the same network architecture?** This is the core principle of experimental design — controlling variables. If we changed both the "world" and the "brain" simultaneously, we couldn't determine whether behavioral differences stem from which factor. Both experiments use the same CfC neuron count, the same AutoNCP wiring strategy, the same 5D emotion output space — differing only in the number and source of perceptual input channels. This way, any behavioral difference can be attributed to "different ecological niches" rather than "different hardware."

```python
#!/usr/bin/env python3
"""
life_wild.py — AI生命体 · 实验A 野生型 (Wild Type)
=====================================================
纯物理传感器驱动 · 无语言模型 · 无叙事层

与实验B (life_v2.py) 的关键差异:
  - 感知来源: Arduino物理传感器 (8通道) 替代 LLM模拟世界 (15通道)
  - CfC输入: 22维 (vs 36维)
  - 移除: LLMPlaywright, Worldbook, EventTranslator
  - 新增: ArduinoSensorBridge, InternalSymbolTracker
  - 保留: CfC情绪核心, 预测引擎, 身体预算, 自主神经, 昼夜节律, 基因组, Phase1-4发育

依赖:  pip install torch ncps
       pip install pyserial  (仅Arduino真实模式)
运行:  python life_wild.py              # 模拟模式
       python life_wild.py --real COM3   # Arduino真实模式
"""

import time, math, random, datetime, os, signal, sys, json, argparse
import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim

# Windows GBK 终端 UTF-8 输出
if hasattr(sys.stdout, 'reconfigure'):
    sys.stdout.reconfigure(encoding='utf-8', errors='replace')
r, 'reconfigure'):
igure(encoding='utf-8', errors='replace')
```

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life_wild.py

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ZIXUN ZHOU 2026年07月21日 18:21

**Step2：emoved Modules — Stripping All Language & Narrative**

Source files (removed from Experiment B):

```
world_v2.py → class LLMPlaywright,
```

```
class EventTranslator; worldbook.py → WORLDBOOK, EVENT_TYPES
```

Three modules completely removed:
LLM Playwright thread (LLMPlaywright),
World Setting Book (Worldbook),
Event Translator (EventTranslator).

**Why "complete deletion" rather than "disabling language output"?** Even without letting the LLM generate language output, as long as world events are still orchestrated by an LLM, information has already been pre-organized by linguistic structure. The LLM-generated sequence "mother enters → breastfeeding → leaves" is essentially a token sequence with causal narrative — even when translated into pure numerical channels, the data still implicitly carries the causal framework of human language. Complete deletion means: the sensory data Experiment A receives contains zero information pre-structured by language. A rise in illumination might be caused by someone turning on a light or clouds dispersing — to Experiment A, this is simply "illuminance value increased," with no conceptual presupposition of "someone," "light switch," or "clouds."

What does this create? A genuinely "pre-linguistic" cognitive entity. If discrete, repeatedly accessible regions spontaneously emerge in its CfC hidden states, these would be "proto-symbols" independent of human language — an extraordinarily rare experimental condition in cognitive science.

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ZIXUN ZHOU 2026年07月21日 18:21

**Step3：New Modules — Sensor Bridge + Internal Symbol Tracker**

Three new modules:
ArduinoSensorBridge,
PhysicalAmbientEngine,
InternalSymbolTracker.

**Why ArduinoSensorBridge?** Real sensors drop packets, have latency, and drift — this isn't a bug, it's a feature. Experiment B's simulated world delivers data with 100% fidelity, but real-world perception is always incomplete. The bridge layer handles these real-world issues (packet loss retry, last-frame fill, drift calibration) while preserving the data's "imperfection." It also supports real/simulated dual-mode switching for debugging without Arduino hardware.

**Why is InternalSymbolTracker the most important new module?** This is the observation instrument for the entire Experiment A. It does three things: ① Online K-Means clustering — continuously monitors whether the CfC's 64-dimensional hidden states spontaneously converge into a few stable "attractor" regions (if they do, these are candidates for "proto-symbols"). ② Transition matrix tracking — if state transitions follow patterns rather than random jumping, the system has formed an internal "grammar" — a sequence structure independent of human language. ③ Symbol stability scoring — a composite 0-1 score combining compactness (within-cluster variance), separation (between-cluster distance), and temporal stability (frame-to-frame consistency), quantifying the degree of internal symbolization.

**Why does this score matter?** It's the key metric for answering the core research question. If symbol_score rises steadily over developmental time, the system is spontaneously developing discrete internal categories without linguistic intervention. If it remains near zero, pure sensory experience without language anchoring is insufficient to produce symbolic representation — which itself would be an important finding.

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```
#!/usr/bin/env python3
"""
sensor_bridge.py — Arduino 物理传感器串口桥
============================================
实验A（野生型）的感知来源。

硬件接线：
  BH1750  (光照)      → I²C (SDA/SCL)
  MAX9814 (声级)      → A0 (ADC)
  DHT22   (温湿度)    → D2 (Digital)
  HC-SR501(人体红外)  → D3 (Digital)
  FSR402  (压力)      → A1 (ADC)
  MPU6050 (加速度计)  → I²C (SDA/SCL)

串口协议：Arduino 每 CYCLE_SEC 秒发送一行 JSON：
  {"lux":320,"db":45,"temp":24.5,"hum":60,"pir":1,"prs":512,"ax":0.1,"ay":-0.02,"

本模块支持两种模式：
  1. 真实模式 (real)  — 从USB串口读取Arduino数据
  2. 模拟模式 (sim)   — 用真实时钟+噪声生成伪传感器数据（无Arduino时调试用）

依赖：pip install pyserial (仅真实模式需要)
"""

import time, math, random, json, sys, os, datetime, threading
from collections import deque

# ═══════════════════════════════════════════════════════════
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空间)
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```

sensor_bridge.py

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```
#!/usr/bin/env python3
"""
symbol_tracker.py — 内部符号涌现监测器
========================================
实验A（野生型）最核心的观测工具。

功能：
  1. 持续记录CfC隐藏状态 h(t) ∈ R^64
  2. 在线K-Means聚类：检测隐藏状态是否自发形成稳定的"吸引子"
  3. 转换频率矩阵：追踪状态之间的转移模式
  4. 符号稳定性指标：量化内部表征的结构化程度

理论依据：
  如果CfC在没有语言介入的情况下，仅通过感官-情绪闭环学习，
  其隐藏状态空间中自发涌现出离散的、可重复访问的区域，
  那么这些区域就是"原始符号"——无需人类语言的内部范畴。

  这是实验A vs 实验B最关键的比较点：
  - 实验B的符号空间被语言锚定（语言是外部范畴化工具）
  - 实验A的符号空间（如果存在的话）完全是内生的
"""

import torch
import numpy as np
import json, os, datetime, threading
from collections import deque

# ═══════════════════════════════════════════════════════════
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Means
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```

symbol_tracker.py

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ZIXUN ZHOU 2026年07月21日 18:21

**Step4: Preserved Core Mechanisms — Comparable Baseline with Experiment B**

All neuroscience mechanisms retained:
sensory preprocessing (habituation/sensitization + developmental gating),
prediction engine,
body budget system,
autonomic nervous controller,
circadian oscillator,
GENOME,
Phase 1-4 developmental timeline.

**Why preserve each mechanism?**

Sensory preprocessing (Groves & Thompson 1970 habituation/sensitization): Attenuation of repeated stimuli and amplification of high-arousal stimuli are fundamental properties of all nervous systems, independent of language. The wild type needs this mechanism to extract signal from noise.

Prediction engine (prediction error drives emotion): Predictive coding is a core operating principle of the brain (Friston's Free Energy Principle), independent of language — even animals without language continuously predict the next frame of sensory input. Prediction error is the driving force for exploration and learning.

Body budget system (Barrett 2017 constructionist emotion theory): Emotions are not "reactions" but byproducts of body budget management. The wild type also needs to manage energy/hydration/thermoregulation, but since there are no "feeding" events, energy recovers through circadian metabolic rhythms — simulating an organism that self-maintains metabolism rather than an infant dependent on caregivers.

Autonomic nervous controller (Porges polyvagal theory): The three-zone hierarchy (freeze/fight-flight/safe-social) is an evolutionarily conserved system present in all vertebrates. The wild type uses PIR infrared signals instead of narrative "social signals" to trigger ventral vagal activation — "something is nearby" rather than "mother is holding me."

GENOME (INxJ temperament parameters): Both experiments use identical innate parameters — same sensory thresholds, arousal baselines, emotional inertia, exploration bias. This ensures any behavioral differences originate from postnatal environment (with language vs. without language) rather than innate temperament. This is the logic of "identical twins raised apart" in experimental design.

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**ZIXUN ZHOU** 2026年07月21日 18:21

**FIELD INVESTGATE - SUWEN YUAN**

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**ZIXUN ZHOU** 2026年07月21日 18:21

**FIELD INVESTGATE - YUYING ZHANG**

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田野调研问卷
你如何“读”一个活体生命
AI Life · RCA Graduate Diploma · Unit 3 Investigate
感谢你参与这项田野调研。它是设计研究项目 AI Life 的一部分，探索人如何感知并联结一个小型的活体生命。填写大约需要 10-15 分钟。你的回答会被匿名使用，仅用于本课程的研究与作品集；你可以随时停止或撤回。
请凭你自己真实的经历来写。如果某道题你确实没有相关经历，直接写“没有”或跳过即可——真实比完整重要。
关于你和你照顾的生物
你的代号（用昵称或任意代称即可，用于匿名）：
Ripple
你照顾的活体生物是（例如：某种植物 / 宠物 / 鱼 / 其他）：
一盆薄荷植物 / 我叫它“小绿灯”
你照顾它多久了：
大约三个月
正文问题
1 | 你在照顾什么活的生物？可以简单介绍一下它。
是什么、你怎么称呼它、它在你生活里大概是什么位置。
我照顾的是一盆放在窗边的薄荷。我叫它“小绿灯”，因为它不是那种很夸张、很戏剧化的植物，但每次长出新叶子时，都会像桌面上亮起一盏很小的绿色信号灯。
它在我生活里的位置有点像一个不会说话的室友：不打扰我，但会让我意识到期间并不是只属于人类。它的存在会提醒我开窗、看光线、摸一摸土，也会让我从自己的事情里短暂抽离出来。对我来说，它不是装饰品，更像是一个需要被持续填期的小生命。
2 | 请回忆最近一次你觉得它“不太对劲”的时候。你是怎么最先察觉的？
尽量具体——第一个引起你注意的细节是什么？是你看到的、听到的、闻到 / 摸到的，还是“就是感觉到了”？你当时是凭直觉，还是先去查了资料？把它当成一个小故事讲出来。
最近一次觉得它“不太对劲”，是有一天早上我发现它的叶子没有像平时那样往窗户方向像散张开，而是有点向下塌，像一把被雨打湿的小伞。最先让我注意到的不是颜色，而是姿态：它平时很“精神”，叶片边缘会轻轻翘起来，那天却像失去了支撑。
我先用手摸了土，发现表面不算特别干，但盆底附近有一点潮，味道也有点闷。我一开始以为它缺水，差点继续浇水，后来查了资料才意识到可能是通风不好加上水分太多。于是我把它移到更通风的位置，松了一点表层土，没有再急着“救它”。过了一天，新叶慢慢抬起来。我觉得那一刻挺有意思：我以为照顾就是马上行动，但它好像在告诉我，有时候不打扰也是一种照顾。
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yuying_zhang.docx

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ZIXUN ZHOU 2026年07月21日 18:21

**FIELD INVESTGATE - QIQI**

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田野调研问卷

你如何“读”一个活体生命

AI Life · RCA Graduate Diploma · Unit 3 Investigate

感谢你参与这项田野调研。它是设计研究项目 AI Life 的一部分，探索人如何感知并联结一个小型的活体生命。填写大约需要 10–15 分钟。你的回答会被匿名使用，仅用于本课程的研究与作品集；你可以随时停止或撤回。

请凭你自己真实的经历来写。如果某道题你确实没有相关经历，直接写“没有”或跳过即可——真实比完整重要。

关于你和你照顾的生物

你的代号（用昵称或任意代称即可，用于匿名）：

七七

你照顾的活体生物是（例如：某种植物 / 宠物 / 鱼 / 其他）：

宠物柴油

你照顾它多久了：

11 年

正文问题

1 | 你在照顾什么活的生物？可以简单介绍一下它。

是什么、你怎么称呼它、它在你生活里大概是什么位置。

我养的一只柴油，我叫他扎西，是藏语里的平安健康的意思。他是我小学五年级考了 100 分，我外公赠送给我的礼物。我觉得它是我主动选择的家人，找它带回家的那一刻，不是买了个宠物，而是签了一份长达十几年的“陪伴契约”。在这个家里，它不需要有什么作用，仅仅因为它是它，就天然拥有一个谁也无法替代的位置。

2 | 请回忆最近一次你觉得它“不太对劲”的时候。你是怎么最先察觉的？

尽量具体——第一个引起你注意的细节是什么？是你看到的、听到的、闻到 / 摸到的，还是“就是感觉到了”？你当时是凭直觉，还是先去查了资料？找它当成一个小故事讲出来。

那天下午，我的小狗照例从阳台的小窝里爬起来，慢吞吞踱到阳光最足的那块地砖上，把自己摊成一张饼。我坐在客厅改稿子，键盘声里混着它偶尔翻身的哼哧声。

后来阳光挪走了，我听见它站起来抖毛的声响，打算从阳台上跳下来找我。可紧接着，落地的声音不对劲：不是平时那种轻巧的“嗵哒”，而是一声闷闷的“咚”，像一袋揽沙子掉在地板上。

我抬头时，它正四脚摆地站在阳台门口，姿势僵着，像被按了暂停键。眼神直直盯着前方，却什么也没在看——不是发呆，是那种大脑空白、身体还没回过神来的懵。足足五秒，它才慢慢眨了一下眼，然后缓缓转过头看我，尾巴轻轻摇了一下，像在说：没事，就是没踩稳。

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ZIXUN ZHOU 2026年07月21日 18:21

**FIELD INVESTIGATE - Z**

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田野调研问卷

你如何“读”一个活体生命

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请凭你自己真实的经历来写。如果某道题你确实没有相关经历，直接写“没有”或跳过即可——真实比完整重要。

关于你和你照顾的生物

你的代号（用昵称或任意代称即可，用于匿名）：

Z

你照顾的活体生物是（例如：某种植物 / 宠物 / 鱼 / 其他）：

狗

你照顾它多久了：

每年两个月不到

正文问题

1 | 你在照顾什么活的生物？可以简单介绍一下它。

是什么、你怎么称呼它、它在你生活里大概是什么位置。

一只萨摩耶，叫香香。寒暑假去我姐家玩，会跟他待一块，晚上带他出去溜溜，夏天一块去吃冰淇淋

2 | 请回忆最近一次你觉得它“不太对劲”的时候。你是怎么最先察觉的？

尽量具体——第一个引起你注意的细节是什么？是你看到的、听到的、闻到 / 摸到的，还是“就是感觉到了”？你当时是凭直觉，还是先去查了资料？找它当成一个小故事讲出来。

姑姑家养了只小中华田园犬，第一次我带他一块去我姑姑家，我进门坐下之后，香香就一直紧贴着我。因为在聊天，所以过了一下才发现他一直在发抖，后来发现他是怕那只小狗。（香香是只大萨摩）

3 | 这种“读懂它状态”的能力，你是一开始就有，还是慢慢养出来的？

现在的你，能读到哪些刚开始养的时候会忽略掉的信号？

偶尔，有时候能猜到他是不是想找人玩或者想让人摸，以及他是不是又想找他妈（我姐）了。

Padlet Drive

Z.docx

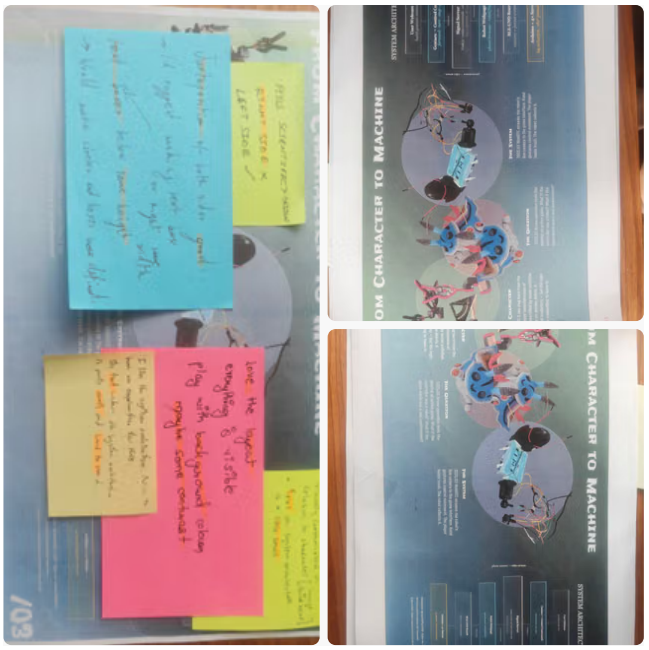
## Portfoilo refelcetion

ZIXUN ZHOU 2026年07月21日 18:21

**Portfolio review session - favourite page**

Image 1: Original layout.
Image 2: Everyone's suggestions.
Image 3: After revision.

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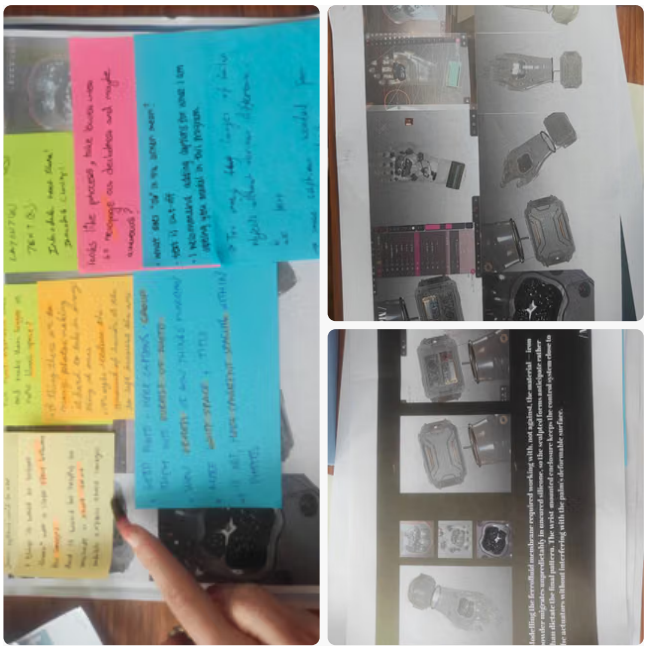
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ZIXUN ZHOU 2026年07月21日 18:21

**Portfolio review session - hardest page**

Image 1: Original layout.
Image 2: Everyone's suggestions.
Image 3: After revision.

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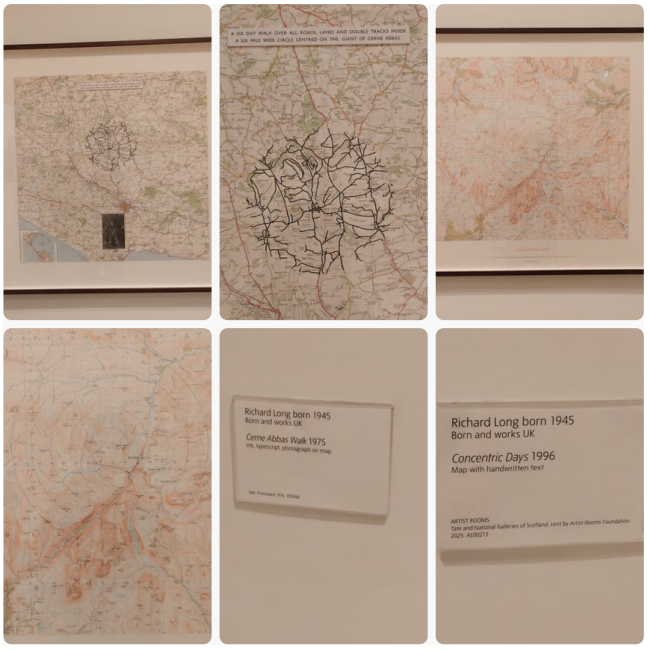


## 版块 5

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Richard Long born 1945
Born and works UK
Cerne Abbas Walk 1975
Richard Long born 1945
Born and works UK
Concentric Days 1996
Map with handwritten text

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